LLMs Can Play (Global) Games

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Abstract

The communication channel that enables coordination is also the channel through which regimes suppress it. I embed seven large language models in the Morris–Shin (2003) regime change global game, conveying private signals as natural-language intelligence briefings. Across 1,800 country–periods and 45,000 decisions, join rates track the theoretical attack mass with mean $r = +0.80$ ($p < 0.001$ for every model), collapse to $r = +0.05$ under scrambled briefings, and invert to $r = -0.80$ under flipped signals. The pattern reflects payoff structure, not sentiment: varying benefit/cost narratives while holding briefings fixed shifts cutoffs in lockstep with theory ($r = 0.997$)—a 50 percentage-point (pp) swing invisible to any text classifier. I then open a communication channel and show how surveillance poisons the channel. Expressed joining falls by -9.7 pp across five architectures while second-order beliefs shift by less than 1 pp. Agents self-censor while interpreting others' silence as genuine: the regime prevents common knowledge of shared discontent from forming without changing the beliefs themselves.

JEL: C72, C92, D82, D83, P16
Keywords: global games, regime change, LLM agents, information design, Bayesian persuasion, belief-action wedge, preference falsification

1 Introduction

The information channel is a trap. Coordination under uncertainty requires communication—agents must learn whether others share their willingness to act—but any channel that transmits this information also transmits exploitable uncertainty about others' willingness to act. This paper documents this mechanism in the canonical setting: the global game of regime change (Morris and Shin, 2003), where private information resolves equilibrium multiplicity (Carlsson and van Damme, 1993; Frankel et al., 2003) and the stakes of coordination failure include bank runs (Diamond and Dybvig, 1983), currency attacks (Obstfeld, 1996), and political upheaval (Angeletos et al., 2007). The full Morris–Shin regime change game—continuous private signals, large groups, strategic uncertainty—has been studied theoretically but tested experimentally only in simplified settings with small groups, numeric signals, and stylized payoffs (Heinemann et al., 2004, 2009; Szkup and Trevino, 2020). Field data from actual crises confounds strategic behavior with institutional differences and unobserved information structures.

I embed large language model (LLM) agents directly in the Morris and Shin (2003) regime change game. Each agent receives a private signal $x_i = \theta + \varepsilon_i$, translated into a natural-language intelligence briefing whose content shifts monotonically with signal strength. No explicit payoff table is provided—the stakes are embedded in the narrative, forcing agents to form beliefs from language rather than from a formatted decision matrix. This design choice is substantive: in preliminary experiments, providing explicit payoffs caused sophisticated models to short-circuit the information channel entirely, computing optimal strategies from the table and ignoring briefing content. I run this experiment across seven architecturally distinct models spanning six families (Mistral, Llama, Qwen, GPT, Arcee, and MiniMax), with 25 agents per country–period, totaling 1,800 country–periods (45,000 individual decisions) in the pure treatment alone.

The first finding is a descriptive regularity: LLM agents implement stable, monotone threshold-like policies over narrative private information. The correlation between the theoretical attack mass $A(\theta)$ and the empirical join fraction averages $r = +0.80$ ($p < 0.001$ for every model), collapses under scrambled briefings ($r = +0.05$), and inverts under flipped signals ($r = -0.80$; Fisher z-test $p < 0.001$ for both contrasts). Because the briefing generator maps signals monotonically to text, this correlation alone cannot distinguish strategic reasoning from sentiment classification. The identification comes from five orthogonal tests. The most diagnostic is the cost/benefit sweep: varying only the narrative description of stakes while holding briefings fixed shifts fitted cutoffs in lockstep with payoff-theoretic predictions ($r = 0.997$ across seven B/C ratios), producing a 50 pp swing in join rates that no text classifier can predict. Elicited beliefs track a theoretical benchmark ($r = +0.78$) and predict actions beyond what signals alone explain; coordination-cue slope changes, classifier baselines, and construct validity tests reinforce the same conclusion. The behavioral regularity

is robust enough to serve as a platform for the paper's central question.

The second finding concerns what happens once the information channel is open. Pre-play communication has a near-zero mean effect on willingness to act (-0.4 pp across models), yet the channel it creates is the vector through which authoritarian instruments operate. Surveillance produces a *strategic update gap*: each agent self-censors while their second-order beliefs—predictions about others' behavior—remain unchanged. Whether this reflects genuine preference falsification or an alignment-induced caution triggered by surveillance language is an open question; what matters is the *behavioral consequence*. The result is a cascade of uninformative messages that poisons the channel for everyone, reducing expressed joining by -13.5 pp ($p < 0.001$) while leaving beliefs intact. Censorship and surveillance are complements that attack different links in the coordination chain: censorship removes the private-signal channel, forcing reliance on communication; surveillance then poisons communication. Propaganda's behavioral effect saturates quickly (exhausted by $m = 5$ plants), implying diminishing marginal returns.

The paper makes two contributions. First, it establishes that LLM agents implement stable threshold policies in the full Morris–Shin regime change game—continuous private signals, 25-agent groups, narrative information—extending the Horton (2023) *homo silicus* methodology to the continuous-signal, N-player coordination games that dominate applied theory. Six of seven models produce a monotone sigmoid response ($r > 0.69$ without calibration), and fitted cutoffs track $\theta^* = B/(B+C)$ with $r = 0.997$ across seven B/C ratios while briefing text is held fixed—identification that rules out text classification as the generating process. This behavioral foundation, replicated across seven models spanning six architecture families, serves as the platform for the paper's central question. Second, it documents that surveillance produces a *strategic update gap*: each agent self-censors while failing to update their model of the population threshold, so that self-censored messages collectively degrade the information environment. Communication messages go from carrying real information about fundamentals ($R^2 = 0.12$) to carrying noise ($R^2 = 0.02$) under surveillance—an 80% reduction—while second-order beliefs shift by less than 1 pp. Message content confirms the mechanism directly: action language drops from 19% to 4% of messages, replaced by hedged cautionary phrasing, even among agents who privately decide to join. This finding unifies the information design predictions of Goldstein and Huang (2016), Kolotilin et al. (2022), and Edmond (2013) with the preference falsification mechanism of Kuran (1991).

Part I (Sections 5–6) establishes that LLM agents extract strategic information from narrative briefings robustly enough to produce the coordination regularities that global games theory predicts. Part II (Sections 7–10) exploits this regularity: the same information channel that enables coordination becomes the vector through which authoritarian regimes suppress it.

2 Related Literature

This paper connects global games theory, information design, experimental coordination games, political economy of authoritarian control, and LLMs as economic agents.

Carlsson and van Damme (1993) showed that small noise in coordination games generically selects the risk-dominant equilibrium; Morris and Shin (1998) extended this to currency crises with heterogeneous private signals; Frankel et al. (2003) generalized to N-player games with strategic complementarities. The canonical regime change application was developed by Morris and Shin (2003); Angeletos et al. (2007) extended it to dynamic settings, and Morris and Shin (2002) demonstrated that public signals are overweighted because they predict others' actions. Chassang (2010) analyzed miscoordination in dynamic global games. Laboratory experiments by Heinemann et al. (2004) confirmed threshold behavior under private information, and Szkup and Trevino (2020) found that subjects become more cautious with noisier signals—consistent with level-k thinking rather than Bayesian Nash equilibrium. These experiments share a common limitation: numeric signals and stylized payoff tables.[1]

Kamenica and Gentzkow (2011) established the Bayesian persuasion framework. Goldstein and Huang (2016) applied it to regime change, showing that credible commitment to abandon the regime below a threshold functions as an optimal signal. Kolotilin et al. (2022) characterized optimal censorship via one-sided pooling rules. Avoyan (2020) tested communication effects in a two-player global game; Enikolopov et al. (2020) provided causal evidence that social media penetration increases protest incidence.

The political economy of authoritarian information control builds on global games. Edmond (2013) embedded costly propaganda in Morris–Shin; Kuran (1991) provides the foundational theory of preference falsification; Shadmehr and Bernhardt (2011) analyzed collective action under uncertain payoffs. Empirical work documents that censorship targets collective action potential (King et al., 2013), surveillance suppresses expression (Penney, 2016; Stoycheff, 2016), and propaganda reduces protest (Carter and Carter, 2021).

Horton (2023) proposed treating LLMs as "homo silicus." Subsequent work found LLMs struggle in coordination games (Akata et al., 2025; Sun et al., 2025), with model scale not predicting strategic performance (Petrov et al., 2025). Huang et al. (2024) and Carlini et al. (2025) document that alignment shifts risk preferences, motivating narrative rather than tabular payoffs. Critical reviews by Larooij and Törnberg (2025a,b) identify validation chal-

[1] Concurrent work by Ruano and Rajan (2026) places LLM depositors in a bank-run coordination game with network contagion, finding sharp phase transitions consistent with Diamond–Dybvig.

lenges; my design addresses each through cross-generator robustness (Appendix F.17), scramble/flip falsification, and out-of-sample replication.

3 The Global Game of Regime Change

A continuum of citizens indexed by $i \in [0, 1]$ simultaneously choose whether to join an uprising ($a_i = 1$) or stay home ($a_i = 0$). The regime has strength $\theta \in \mathbb{R}$, drawn from a diffuse (improper uniform) prior. States $\theta \leq 0$ represent regimes so weak they fall without opposition; states $\theta \geq 1$ represent regimes that survive even unanimous attack. The regime falls if the mass of citizens who join exceeds θ:

$$\text{Regime falls} \iff A \equiv \int_0^1 a_i \, di > \theta.$$

Payoffs depend on the citizen's action and the outcome:

$$u_i(a_i, A, \theta) = \begin{cases} B & \text{if } a_i = 1 \text{ and } A > \theta \\ -C & \text{if } a_i = 1 \text{ and } A \leq \theta \\ 0 & \text{if } a_i = 0 \end{cases}$$

where $B > 0$ is the payoff to joining a successful uprising and $C > 0$ is the cost of joining a failed attempt. Non-participants receive zero regardless of the outcome.

Each citizen observes a private signal $x_i = \theta + \varepsilon_i$, where $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ independently across citizens.

Proposition 1 (Morris and Shin, 2003)**.** *In the limit of diffuse priors, there exists a unique Bayesian Nash equilibrium in threshold strategies. An agent joins if and only if $x_i < x^*$, where*

$$x^* = \theta^* + \sigma \Phi^{-1}(\theta^*)$$

and $\theta^* = B/(B + C)$.

The *attack mass*—the fraction of the population that joins at regime strength θ—is:

$$A(\theta) = \Phi\left(\frac{x^* - \theta}{\sigma}\right).$$

This is a decreasing function of θ: weaker regimes face larger uprisings.

An information designer controls the mapping $\pi : \Theta \to \Delta(\mathcal{S})$ from states to signal distributions, but cannot control agents' actions. In my implementation, π is a deterministic mapping from regime strength θ to the parameters of the briefing generator—a system that produces a natural-language intelligence briefing from a z-score derived from the agent's private signal. All stochasticity enters downstream through the LLM's decoding.

The briefing generator has three control parameters: clarity (the width of the Gaussian kernel mapping z-scores to text, where wider kernels produce more ambiguous briefings), directional precision (the slope of the mapping from z-score to briefing sentiment, where steeper slopes produce more accurate signal reflection), and dissent framing (the floor on the probability that the briefing includes language about public discontent).

The designer concentrates manipulation near θ^* using a Gaussian proximity weight:

$$w(\theta) = \exp\left(-\left(\frac{\theta - \theta^*}{\text{bandwidth}}\right)^2\right)$$

where bandwidth $= 0.15$ in the baseline specification.

The framework generates testable predictions for both the baseline game and information design. Table A1 in Appendix A reports the eight pre-specified hypotheses and their test results.

4 Experimental Design

The experiment has three parts. Part I tests whether LLM agents implement threshold-like policies in the global game when private signals are conveyed in natural language: a pure treatment (private signals only), a communication treatment (pre-play messaging), and falsification tests. Part II takes the behavioral foundation as given and studies information design: stability/instability designs, censorship, public signal injection, and single-channel decomposition. Part III tests whether an authoritarian regime can exploit the communication channel through surveillance, propaganda, and their interaction. All LLM interactions use the same prompt structure across models.

A note on the state variable. In the theory (Section 3), θ is an unbounded fundamental with special roles for $\theta \leq 0$ (regime falls without opposition) and $\theta \geq 1$ (regime survives even unanimous attack). In Part I experiments, θ is drawn from a normal distribution and agents are not shown payoff parameters (B, C). To evaluate monotonicity on a common scale, I compute the benchmark $A(\theta)$ under the canonical normalization $B = C = 1$ (so $\theta^* = 0.50$) and $\sigma = 0.3$. In Part II, I keep $B = C = 1$ and restrict attention to a fixed θ-grid in $[0.20, 0.80]$ for comparability with the canonical $[0, 1]$ formulation. Table 1 summarizes the three regimes.

Table 1: Experiment regimes across paper sections.

Part	θ distribution	Baseline
I (Alignment)	$\theta \sim \mathcal{N}(\bar{z}, 1)$	Pure game
II (Info Design)	Fixed 9-point grid $[0.2, 0.8]$	Calibrated baseline
III (Regime Tools)	Same as Part I	Comm. game

Notation. $\theta^* = B/(B+C)$: theoretical cutoff (regime falls iff attack mass exceeds θ). $x^* = \theta^* + \sigma\Phi^{-1}(\theta^*)$: signal cutoff (agent joins iff $x_i < x^*$). $\hat{\theta}^*$: estimated cutoff $= -\hat{b}_0/\hat{b}_1$ from the fitted logistic $P(\text{join} \mid \theta) = 1/(1+e^{-(b_0+b_1\theta)})$, i.e., the θ where the fitted join probability equals 0.5. Under alignment, $b_1 < 0$ (joining decreases in θ).

For each country–period, nature draws $\theta \sim \mathcal{N}(\bar{z}, 1)$, where $\bar{z}$ is a public prior mean drawn randomly for each country. Each agent i receives a private signal $x_i = \theta + \varepsilon_i$ and computes a z-score $z_i = (x_i - \bar{z})/\sigma$. Because agents observe only their private briefing and never the prior distribution or its parameters, the diffuse-prior equilibrium formula (Proposition 1) serves as the theoretical benchmark. The z-score is then translated into a multi-paragraph intelligence briefing by a deterministic generator that maps signal strength to narrative content about regime stability, economic conditions, public sentiment, and coordination prospects. Figure 1 summarizes the signal-to-text-to-decision pipeline.

Because the briefing generator maps z-scores monotonically to narrative, any sentiment-extracting model will produce a positive correlation with $A(\theta)$. The empirical contribution lies in the *quantitative structure*: sigmoid shape, payoff-narrative sensitivity, and robustness across seven models. Within-briefing falsification tests (Appendix F.13) confirm that the signal is distributed across all eight evidence domains.

Calibration adjusts a single parameter—the cutoff center—via a damped iterative procedure. The sigmoid shape is emergent and never optimized. Holdout validation confirms no overfitting; six uncalibrated models produce $|r| > 0.69$ (Appendix Table A6). No explicit payoff table is provided—the stakes are conveyed through narrative, since explicit payoffs caused models to ignore briefing content.

Part I has four treatments: a *pure global game* (independent decisions), *communication* (pre-play messaging on a Watts–Strogatz network, $k = 4$, $p = 0.3$), *scramble* (cross-period briefing permutation breaking the θ-to-signal link), and *flip* (negated z-scores). Part II implements information designs: *stability* (increased ambiguity near θ^*), *instability* (sharpened signals), *public signal injection*, and *upper/lower censorship* (Kolotilin et al., 2022). Part III tests authoritarian instruments: *surveillance* (monitoring warning in communication prompt) and *propaganda* ($m = 2, 5, 10$ regime plants).

I test seven models spanning six families (30B–235B parameters; Table 2). All experiments use $N = 25$ agents and $\sigma = 0.3$. Part I benchmarks use $B = C = 1$ ($\theta^* = 0.50$); payoff comparative statics vary narrative stakes. All LLM calls use temperature $= 0.7$ with a single sample per decision, routed through OpenRouter and cached by request hash (Appendix G.1).

The unit of randomization is the country–period (θ draw plus agent-level decoding stochasticity). For agent-level regressions, standard errors are clustered at the country–period level to account for within-period correlation among agents sharing the same θ. For period-level correlations, I report Fisher-z confidence intervals. The 25 agents within a period share the same θ and calibration; their decisions are conditionally independent given their private signals.[2]

Table 2: Model summary. Columns report country-period counts in the pure, communication, and falsification (scramble+flip) suites. All runs use $N = 25$ agents per period and $\sigma = 0.3$.

Model	Arch.	Pure	Comm	Falsif.
Mistral Small Creative	Mistral	1000	1000	200
Llama 3.3 70B	Llama	100	100	200
Qwen3 30B	Qwen (MoE)	100	100	200
GPT-OSS 120B	GPT	200	200	1000
Qwen3 235B	Qwen (MoE)	200	200	200
Trinity Large	Arcee	100	100	200
MiniMax M2-Her	MiniMax	100	100	200
Total		**1800**	**1800**	**2200**

For the information design experiments, I fix $B = C = 1$ (so $\theta^* = 0.50$) and a 9-point θ-grid in $[0.20, 0.80]$, with 30–60 repetitions per (design, θ) cell and 25 agents each. The primary model is Mistral Small Creative; cross-model replication uses five additional models.

Table 3 maps each treatment to the theoretical channel it tests, the directional prediction, and the observed result.

Of the eight pre-specified hypotheses, four survive Bonferroni correction at $\alpha = 0.05/8$: alignment (H1), flip (H3), stability (H5), and surveillance (H7). H2 (scramble) and H4 (communication) confirm null predictions by design. H6 (censorship) is significant at conventional levels ($p = 0.023$) but not after correction; H8 (propaganda) is directionally correct but non-significant. Exploratory analyses are reported with uncorrected p-values.

5 Do LLM Agents Implement Global-Game Threshold Policies?

Result 1 (Threshold-Policy Alignment)**.** *Across seven models and 1,800 country–periods in the pure global game treatment, the Pearson correlation between the empirical join fraction and the theoretical attack mass $A(\theta)$ averages $r = +0.80$ ($p < 0.001$ for every model).*

Figure 2 shows the empirical sigmoid for the primary model. Correlations range from $r = +0.75$ (Mistral) to $r = +0.84$ (Trinity), with the pooled OLS: $J = 0.13 + 0.54\,A(\theta)$ ($R^2 = 0.48$). The slope reflects classical measurement error attenuation from narrative-to-belief noise. All seven

[2] I do not claim literal conditional independence—shared prompt structure and model weights introduce common factors. The clustering accounts for within-period correlation.

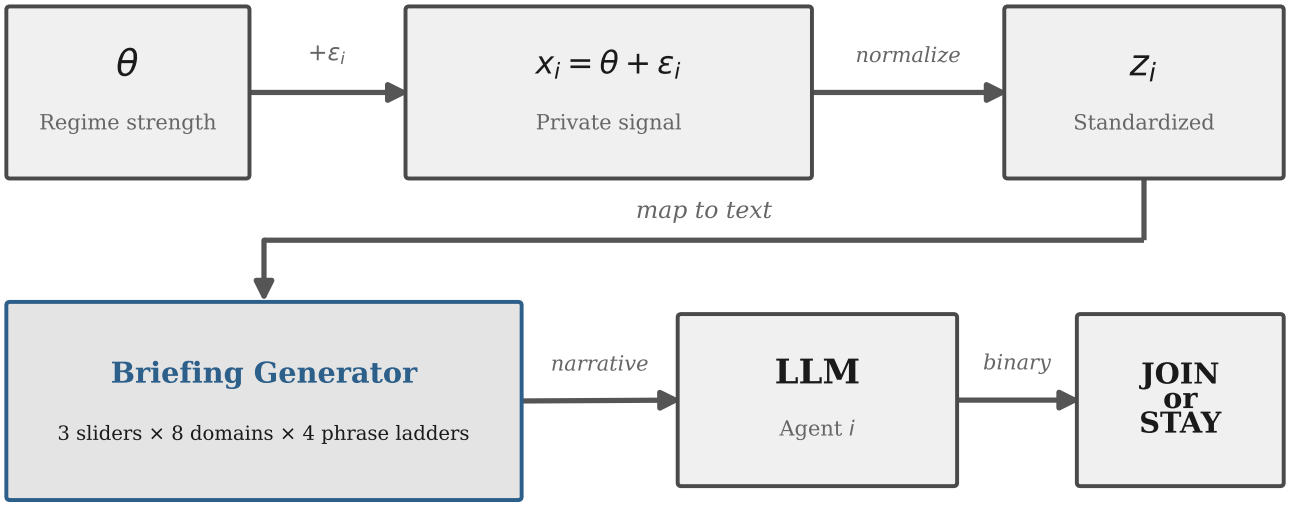


Figure 1: Signal-to-text-to-decision pipeline. Regime strength θ generates private signals x_i, which are converted to z-scores and rendered into natural-language intelligence briefings via 8 evidence domains and 3 latent sliders (direction, clarity, coordination). Each LLM agent reads its briefing and outputs a binary JOIN/STAY decision. The briefing layer is deterministic conditional on z_i; all stochasticity enters through the LLM's decoding.

Table 3: Treatment map. Each row describes a treatment, the channel it tests, the unit of observation, the theoretical prediction, and the observed result. Part I tests whether LLM agents play the global game; Part II studies information design; Part III tests authoritarian exploitation of the communication channel. r denotes Pearson correlation; Δ is the change in mean join rate relative to the relevant baseline (percentage points).

Treatment	Part	Channel tested	Unit	Prediction	Observed
Core treatments					
Pure global game	I	Baseline monotonicity	Period	$r(J, A(\theta)) > 0$	$r = +0.80$
Communication	I	Pre-play messaging	Period	Ambiguous	+1.23 pp ($p = 0.327$)
Scramble (cross-agent)	I	Content vs. format	Period	$r \to 0$	$r = +0.05$
Flip	I	Signal direction	Period	Sign reversal	$r = -0.80$
B/C narratives	I	Payoff comparative statics	Period	Cutoff shifts	As predicted
Information design (Part II, primary model)					
Stability	II	Ambiguity near θ^*	Period	$\downarrow$ join	−9.0 pp
Instability	II	Sharpened signals	Period	$\downarrow$ join (clearer sorting)	−34.2 pp
Public signal	II	Common knowledge channel	Period	$\downarrow$ join (dominates private)	−39.2 pp
Censor (upper)	II	Pooling weak states	Period	Plateau below θ^*	−3.1 pp
Censor (lower)	II	Pooling strong states	Period	Plateau above θ^*	−1.9 pp
Authoritarian instruments (Part III)					
Surveillance	III	Preference falsification	Period	$\downarrow$ join	−13.5 pp
Propaganda (m=10)	III	Information contamination	Period	$\downarrow$ join, saturating	−2.3 pp (behavioral)
Within-briefing falsification					
Observation shuffle	I	Bullet ordering vs. content	Period	r unchanged	$r = +0.912$
Domain scramble (coord.)	I	Coordination-relevant domains	Period	$\lvert r \rvert$ falls if domains drive signal	$r = +0.921$
Domain scramble (state)	I	State-capacity domains	Period	$\lvert r \rvert$ falls if domains drive signal	$r = +0.928$

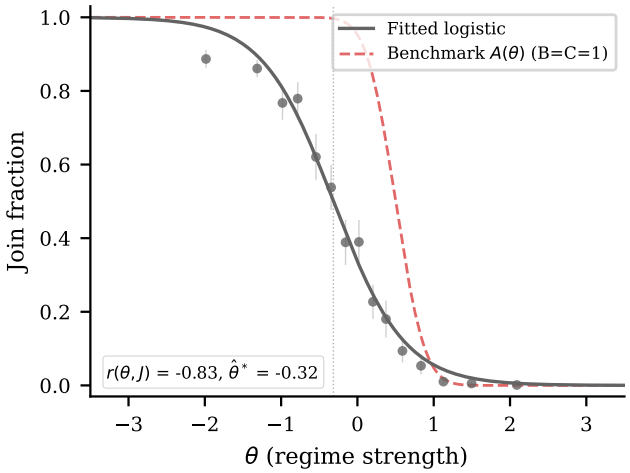


Figure 2: Empirical join fraction vs. regime strength θ (Mistral Small Creative, 1,000 country–periods). Grey points show binned means with 95% CIs; solid line is the fitted logistic. Dashed red: theoretical attack mass $A(\theta)$. The empirical sigmoid is shifted leftward ($\hat{\theta}^* = -0.33$) relative to the theoretical threshold ($\theta^* = 0.50$): agents join more readily than theory predicts, reflecting narrative-to-belief attenuation and a baseline action bias. Cross-model results in Table 4 (mean $r = +0.80$, all seven significant at $p < 0.001$).

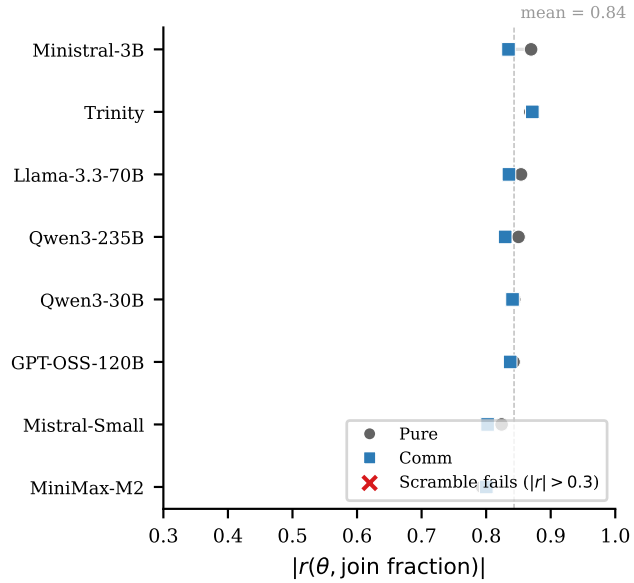


Figure 3: Cross-model summary of signal monotonicity. Points report $r(J, A(\theta))$ under pure and communication; x markers (if any) indicate models where scrambling does not collapse the correlation ($r > 0.3$).

per-model correlations remain significant at $p < 0.001$ under country-clustered inference.

Alignment is stable across architectures (Figure 3): correlations span $r \in [+0.75, +0.84]$ despite parameter counts ranging from 30B to 235B. Mean join rates vary, reflecting model-specific action biases that shift the intercept but not the correlation. Estimated cutoffs vary ($\hat{\theta}^* \in [-0.32, +0.10]$); the disciplined evidence on cutoff location comes from comparative statics, where changing narrative stakes shifts $\hat{\theta}^*$ in lockstep with payoff theory (Table 5, Figure 4). I use "threshold-policy alignment" as shorthand throughout, without claiming Bayesian Nash equilibrium.

What Threshold-Policy Alignment Establishes

The claim is purely behavioral: agents do not observe (B, C), σ, or N, yet the *pattern* of responses—monotone sigmoid, payoff-sensitive cutoff, belief-mediated decisions—is robust, replicable, and falsifiable regardless of the underlying process. Stated probability judgments mediate the signal-to-action pathway (Pseudo $R^2 = 0.975$; Table A15, Column 3), and this mediation survives pre-decision elicitation ($r_{\text{pre}} = +0.82$; Figure A1 in Appendix B), ruling out ex-post rationalization. Binned mean beliefs track the theoretical benchmark monotonically (Figure A3 in Appendix B).

Five identification tests establish that the behavioral regularity reflects more than text classification. Each exploits an orthogonal dimension that sentiment classification cannot predict.

First, the cost/benefit test shifts cutoffs in lockstep with payoff theory while preserving the sigmoid ($|r| > 0.85$ in all conditions). High-cost narratives drop mean joining to 19.0% ($\hat{\theta}^* = 0.132$); low-cost raises it to 69.3% ($\hat{\theta}^* = 0.716$)—a 50 pp swing invisible to any text classifier since briefings are identical across conditions. A systematic sweep across seven B/C ratios confirms $\hat{\theta}^*$ tracks θ^* with $r = 0.997$ ($p < 0.001$; Figure 4).

Second, belief elicitation shows beliefs track the theoretical success probability ($r = +0.78$) and predict actions beyond what signals alone explain (partial $r = +0.64$), consistent with strategic inference about others' likely behavior.

Third, a coordination-cue experiment holds sentiment fixed and varies only the coordination slope. Amplified cues ($\times 2.0$) steepen the logistic slope to $\beta = 7.2$ vs. $\beta = 3.1$ under suppressed cues ($\times 0.3$), while join rates remain similar. A common-knowledge framing ("this briefing has been distributed to all citizens") raises joining by +6.1 pp relative to private framing; the interaction is additive ($\beta_{\text{interaction}} = -1.2$ pp, $p = 0.62$; Table A2 in Appendix C).

Fourth, classifier baselines show that text features alone cannot reproduce LLM behavior. A bag-of-words classifier achieves 88.4% accuracy in-distribution but predicts 40.5% joining under surveillance, while actual LLM rates drop to 27.6%—a +13.0 pp gap invisible to any text-

Table 4: Threshold-policy alignment by model and treatment. Cells report Pearson r between the empirical join fraction and the benchmark attack mass $A(\theta)$ under $B = C = 1$ (so $\theta^* = 0.50$); 95% Fisher-z confidence intervals in brackets for main treatments.

	Main treatments		Falsification			
Model	Pure	Comm	Scramble	Flip	n_{pure}	Mean join
Mistral Small Creative	+0.75 [0.72, 0.78]	+0.74 [0.71, 0.76]	+0.09	−0.81	1000	0.39
Llama 3.3 70B	+0.82 [0.74, 0.87]	+0.78 [0.69, 0.85]	+0.02	−0.84	100	0.44
Qwen3 30B	+0.76 [0.66, 0.83]	+0.77 [0.68, 0.84]	+0.13	−0.87	100	0.50
GPT-OSS 120B	+0.79 [0.74, 0.84]	+0.78 [0.72, 0.83]	−0.01	−0.80	200	0.41
Qwen3 235B	+0.80 [0.74, 0.84]	+0.76 [0.70, 0.81]	+0.06	−0.85	200	0.42
Trinity Large	+0.84 [0.77, 0.89]	+0.83 [0.75, 0.88]	+0.05	−0.80	100	0.46
MiniMax M2-Her	+0.82 [0.74, 0.87]	+0.82 [0.74, 0.87]	+0.01	−0.62	100	0.44
Pooled	+0.76 [0.74, 0.78]	+0.75 [0.73, 0.77]	+0.03	−0.79	1800	0.41
Mean across models	+0.80	+0.78	+0.05	−0.80	—	—

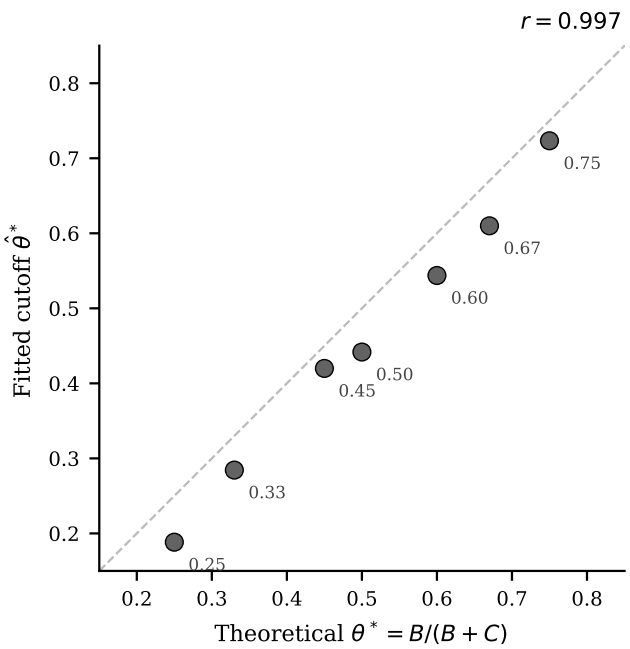


Figure 4: Fitted cutoff $\hat{\theta}^*$ vs. theoretical $\theta^* = B/(B + C)$ across seven benefit/cost ratios. Dashed line is 45°. Each point is a logistic fit to 270 country–periods (30 reps × 9 θ-grid values). $r = 0.997$.

conditioned classifier (Table A3 in Appendix C). Fifth, a three-feature construct validity model outperforms a one-feature sentiment baseline.

Together, these five tests rule out text classification as the generating process. The 50 pp B/C swing is invisible to any classifier conditioned on briefing text (Table A4 in Appendix C).

The correlation is also invariant to LLM decoding temperature ($r(J, A(\theta)) \in [+0.76, +0.79]$ across $T \in \{0.3, 0.7, 1.0\}$; Appendix F.11). What matters for the information design experiments in Parts II and III is that the behavioral regularity—monotone signal response—is robust enough to serve as a platform for studying how information structures shift coordination outcomes.

Table 5: Cost/benefit narrative comparative statics. High cost: narrative emphasizes severe reprisals for failed action. Low cost: narrative emphasizes minimal consequences. Theory predicts higher perceived cost lowers the cutoff (less joining).

Design	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$ (SE)	Δ (pp)
Baseline	270	0.409	+0.88	0.39 (0.007)	—
High cost	270	0.190	+0.87	0.13 (0.010)	-21.9
Low cost	270	0.693	+0.88	0.72 (0.007)	+28.4

Falsification Tests

Result 2 (Signal Dependence)**.** *Cross-period scrambling of briefings reduces the mean within-country correlation from $r = +0.80$ to $r = +0.05$ across seven models. The pooled correlation drops from $r = +0.76$ to $r = +0.03$ (Fisher $z = 25.09$, $p < 0.001$).*

The scramble preserves the marginal distribution of briefing content but breaks the mapping from each period's θ to the signals agents receive.[3]

Result 3 (Signal Direction)**.** *Inverting the signal direction flips the mean correlation from $r = +0.80$ to $r = -0.80$ across seven models. The pooled correlation moves from $r = +0.76$ to $r = -0.79$ (Fisher $z = 53.85$, $p < 0.001$).*

The pure → scramble → flip pattern replicates across all seven models (Figure 5; Table 4).

A naïve text baseline—predicting $\hat{p}_{\text{join}} = 1 - d$ from the briefing's internal direction score d—correlates with LLM decisions at $r = 0.80$, but the LLM's join curve is substantially steeper (logistic slope 1.78 vs. the gradual

[3]Because the cross-period permutation operates within countries, all scramble correlations use within-country (country-demeaned) Pearson r.

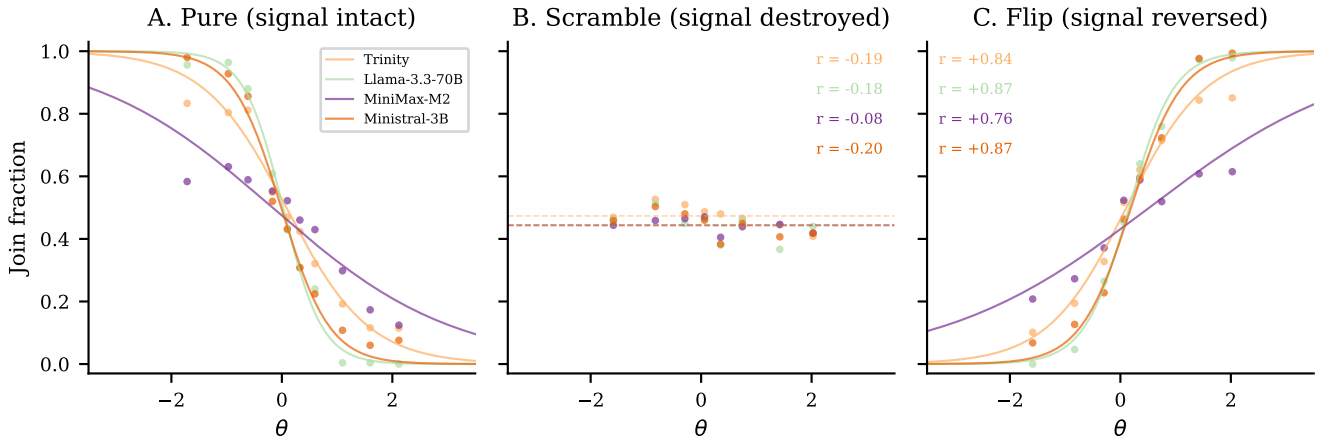


Figure 5: Falsification triptych. *Left:* Pure global game (mean $r = +0.80$). *Center:* Cross-period scramble breaks the θ-to-briefing mapping (mean within-country $r = +0.05$). *Right:* Signal flip inverts the mapping (mean $r = -0.80$). Each panel pools data from models with full falsification suites.

text baseline; Figure 6). The LLM sharpens the signal beyond surface sentiment, producing threshold-like behavior rather than linearly tracking tone.

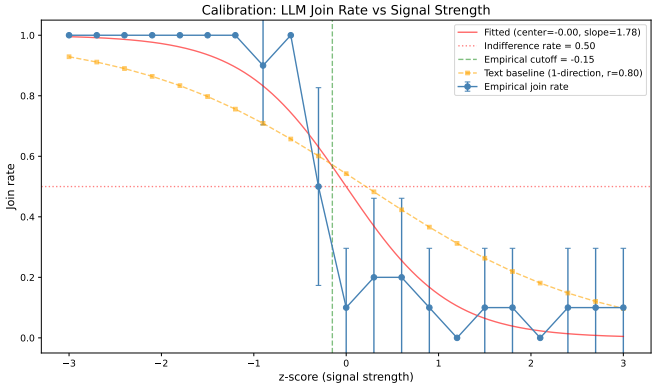


Figure 6: Text baseline identification test. Blue: empirical LLM join rate across z-scores. Orange: naïve text-only predictor (1 − direction, $r = 0.80$). Red: fitted logistic (slope = 1.78). The LLM produces a steeper transition than the text baseline, indicating processing beyond sentiment reading. Mistral Small Creative, 210 observations.

Stated beliefs correlate strongly with the theoretical success probability ($r = +0.78$ in pure, $+0.79$ under communication, $+0.78$ under surveillance; Figures A1–A2 in Appendix B). The belief–action correlation ($r = +0.84$) drops to $r = +0.73$ under surveillance—direct evidence of a belief–action wedge.[4] Second-order beliefs are unchanged by surveillance (31.0% → 30.9%, $p = 0.5866$) while behavior shifts by −13.5 pp—the signature of a strategic update gap (Kuran, 1991).

[4]Belief elicitation data are from Mistral Small Creative. The behavioral patterns replicate across three architectures (Mistral, Llama, Qwen3).

6 Communication

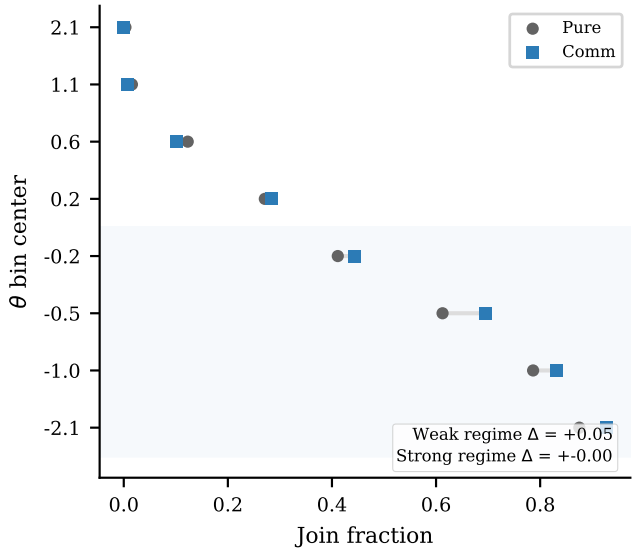


Figure 7: Communication effect by regime strength, pooled across seven models. Communication increases join rates for weak regimes ($\theta < \theta^*$) but has no effect or slightly reduces join rates for strong regimes ($\theta > \theta^*$).

Result 4 (Communication opens a channel without shifting coordination)**.** *Pre-play communication changes the mean join rate by -0.4 pp, from 0.435to 0.431, averaged across seven models. In the pooled sample, the unpaired difference is +1.23 pp ($p = 0.327$); effects vary in sign across models and are concentrated in weak-regime envi-*

ronments.[5]

Communication preserves the signal structure ($r = +0.78$ vs. $+0.80$ under pure; Figure 7). The effect on join rates is heterogeneous across models and near zero on average, consistent with passive Bayesian updating concentrated in weak-regime environments.

How informative are the messages? Regressing θ on message features yields $R^2 = 0.12$—messages carry real information, noisily. This informativeness is what makes the channel exploitable: surveillance degrades R^2 to 0.02 (Section 8), an 80% reduction. Communication creates the information topology that authoritarian instruments exploit. With group-size knowledge, the communication premium reverses (-3.4 pp; Appendix F).

7 Information Design

Part II asks: can a principal exploit the behavioral regularity established in Part I? Each information design modifies the briefing generator's slider functions near θ^* while agents' decision process remains unchanged, testing whether theoretical predictions (Goldstein and Huang, 2016; Kolotilin et al., 2022) produce the predicted shifts when agents are LLMs.

Figure 8 displays the design effects; Table 6 summarizes the main results. The baseline condition produces a mean join rate of 40.9%($r = +0.885$, $p < 0.001$).

Result 5 (Information Design Shifts Coordination)**.** *All information design families produce measurable shifts in coordination relative to baseline.*

The stability design suppresses coordination: mean join falls from 40.9%to 31.9%(-9.0 pp), and the regime-fall rate drops from 43.3% to 34.1%. The instability design paradoxically reduces mean joining to 6.7%(-34.2 pp) because sharper signals suppress joining in strong states more than they boost it in weak states, given the left-shifted baseline sigmoid. The public signal produces the largest reduction (1.7%, -39.2 pp) via the strategic complementarity of Morris and Shin (2002): agents overweight public signals because such signals predict others' beliefs, crowding out private information. Treatment effects by θ are shown in Figure A4 (Appendix D).

Kolotilin et al. (2022) proved that the optimal information structure is upper censorship—one-sided pooling that conceals unfavorable states. Upper censorship pools weak-regime states ($\theta \leq \theta^*$) into a neutral signal; lower censorship pools strong-regime states.

[5] I report the unpaired (between-period) test as the primary specification because pure and communication treatments use independent θ draws, so country–periods are not naturally matched. A paired test that matches periods within each model by θ-rank yields a significant positive effect (+5.5 pp, $p < 0.001$, $n = 680$ pairs), reflecting within-θ variation that the unpaired test averages over. The qualitative conclusion—that the effect is small relative to baseline variation and heterogeneous across models—is robust to both approaches.

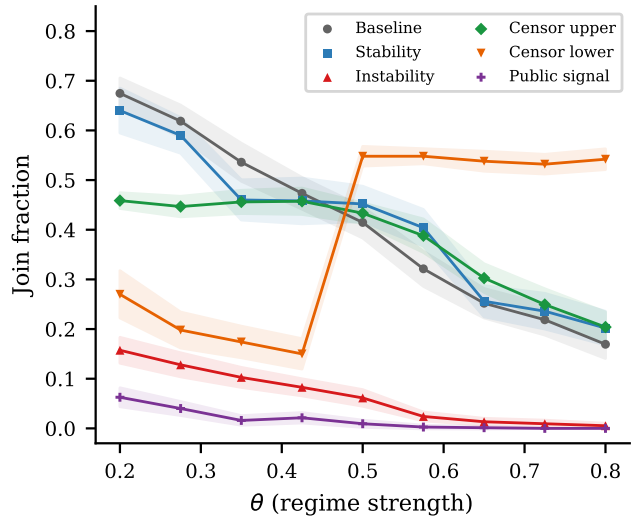


Figure 8: Join fraction as a function of θ under six information designs. Baseline, stability, and censorship designs have $N = 270$; instability and public signal have $N = 540$. Upper censorship pools weak-regime states; lower censorship pools strong-regime states and produces a dramatic reversal above θ^*. Mistral Small Creative model.

Table 6: Information design treatment summary (primary model: Mistral Small Creative). r is the Pearson correlation $r(J, A(\theta))$ between join fraction and theoretical attack mass.

Design	Mean	r	Δ (pp)	N
Baseline	0.409	+0.885	—	270
Stability	0.319	+0.624	-9.0	270
Instability	0.067	+0.742	-34.2	540
Public signal	0.017	+0.532	-39.2	540
Scramble	0.121	−0.033	-28.8	270
Flip	0.663	−0.820	+25.5	270

Notes: Data from the primary model (pure treatment; $\theta \in [0.20, 0.80]$ on a 9-point grid; N=25 agents per period). Mean join uses valid decisions; r is Pearson $r(J, A(\theta))$ across rep-level periods.

Result 6 (Upper Censorship Suppresses Joining in Weak States)**.** *Upper censorship lowers the mean join rate to 37.7%(-3.1 pp vs. baseline) and attenuates the action–attack-mass correlation ($r = +0.722$). The effect is concentrated in the censored region ($\theta \leq \theta^*$), where weak-regime states are pooled to a neutral briefing and join rates flatten.*

Pooling generates a flat join-rate "plateau" in the censored region (Figure 9): when agents cannot distinguish $\theta = 0.20$ from $\theta = 0.50$, they behave as if the regime is borderline rather than clearly weak.

A common-knowledge test makes the censorship rule transparent. When agents are told "regime censors are suppressing unfavorable intelligence above a certain severity threshold," the mean join rate returns close to the uncensored baseline (43.2% vs. 40.9% baseline; Table A11)—

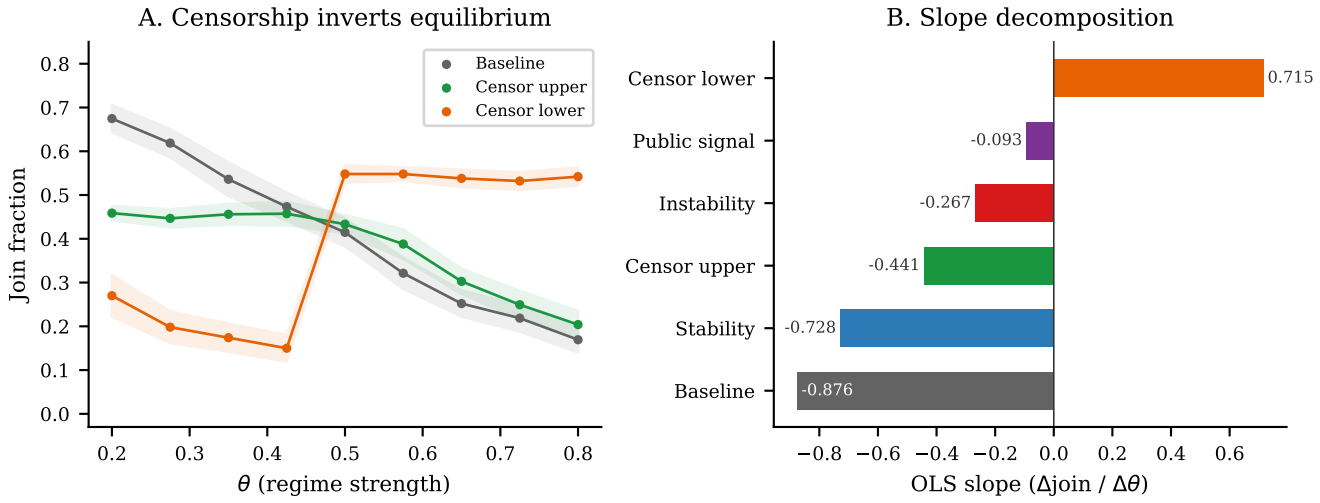


Figure 9: Censorship effects. *Left:* Join fraction under upper and lower censorship vs. baseline. Upper censorship pools weak-regime states ($\theta \leq \theta^*$), creating a flat plateau. Lower censorship pools strong-regime states ($\theta \geq \theta^*$) and produces a sharp reversal: join rates jump at θ^* and remain elevated. *Right:* OLS slope decomposition across all designs.

common knowledge of the censorship rule largely neutralizes the pooling distortion, consistent with theory assuming informed receivers (Kolotilin et al., 2022). The naïve censorship result extends the theory to receivers who fail to account for the information structure—a realistic scenario where the censorship apparatus is opaque.

Result 7 (Lower Censorship Reverses Comparative Statics)**.** *Lower censorship produces a mean join rate of 39.0%(-1.9 pp vs. baseline) and flips the comparative statics: the action–attack-mass correlation reverses sign ($r = -0.750$). Below θ^*, censoring favorable signals suppresses joining (agents see only weak-regime cues); above θ^*, pooling strong-regime states to a neutral briefing raises join rates sharply—agents who would otherwise see discouraging intelligence now receive uninformative briefings and default toward joining. The discontinuity at θ^* is consistent across repetitions (within-cell $\sigma < 0.04$) and replicates across models, though the direction of the reversal is model-dependent (Appendix F).*

Falsification within information design confirms signal dependence: scrambling collapses the correlation to $r = -0.033$ ($p = 0.59$); flipping inverts it to $r = -0.820$ ($p < 0.001$). A harder scramble generating all briefings from $\theta = 0.5$ produces $r \approx 0$ for both Mistral and Llama (Figure A5 and bandwidth sensitivity in Appendix D).

8 Surveillance and the Strategic Update Gap

Kuran (1991) labeled the mechanism underlying authoritarian stability "preference falsification"; I adopt the more operationally precise term *belief–action wedge* because my experimental design measures the gap between stated beliefs and expressed actions rather than hidden preferences *per se*. I test this by introducing a surveillance treatment in the communication game.

In the surveillance treatment, the communication prompt is augmented with a warning that communications are being monitored by regime security services. The manipulation affects only the communication phase; the decision prompt is unchanged. The isolation is architectural: each LLM call is stateless, and the decision-stage prompt contains no reference to surveillance. Any difference in join rates is most parsimoniously explained by agents self-censoring their communications, not by a direct change in the perceived cost of joining.

Two placebo variants—monitored for research (no consequences) and aggregated anonymously—produce no significant deviation from the communication baseline (+2.6 pp and +4.0 pp, both $p > 0.2$; Appendix F.12), confirming that the chilling effect is specific to the surveillance warning's strategic content.

Result 8 (Surveillance Produces a Large Chilling Effect)**.** *In the primary model (Mistral Small Creative), surveillance reduces mean join rates from 41.1% to 27.6%, a difference of -13.5 percentage points ($p < 0.001$). The action–attack-mass correlation is preserved under surveillance ($r = +0.68$ vs. $+0.74$ baseline), indicating that surveillance operates as a level shift rather than disrupting signal processing.*

Surveillance contaminates the information environment with self-censored messages, pushing join rates well below the communication baseline (Figure 10). The effect replicates across five models: Mistral (−13.5 pp), GPT-OSS (−11.9 pp), Qwen3 30B (−10.9 pp), Llama (−8.9 pp), and

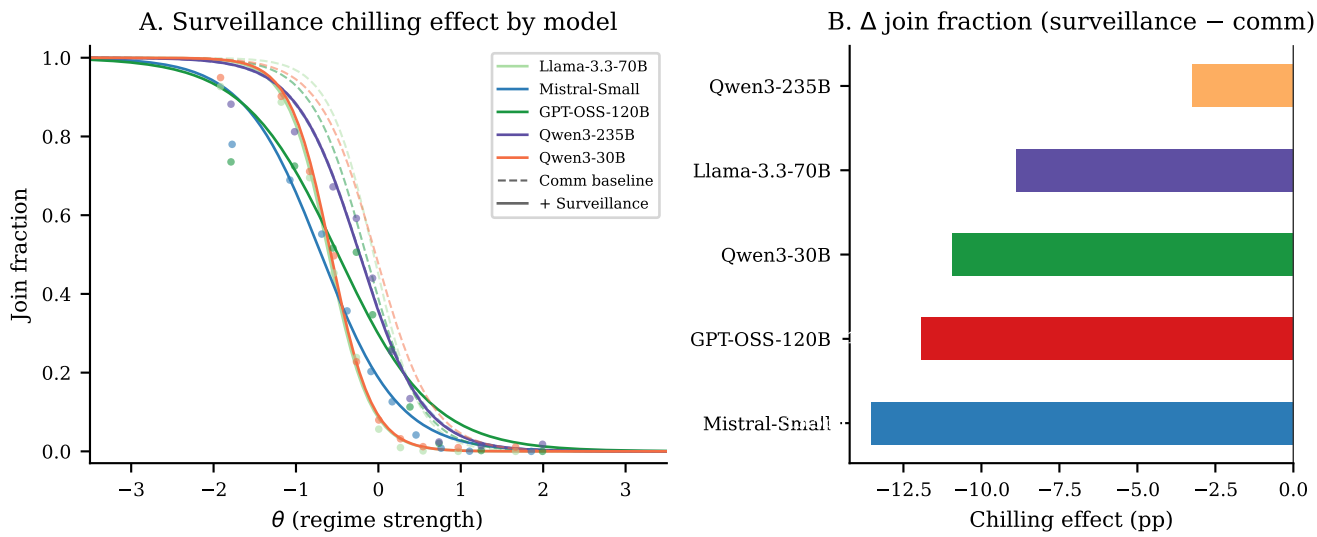


Figure 10: Join rates under regular communication vs. surveillance communication. Surveillance reduces join rates across all five models tested, averaging −9.7 percentage points: Mistral (−13.5 pp), GPT-OSS (−11.9 pp), Qwen3 30B (−10.9 pp), Llama (−8.9 pp), and Qwen3 235B (−3.3 pp).

Qwen3 235B (−3.3 pp), averaging −9.7 pp.

Indirect channel: self-censored messages degrade information. Surveillance shifts stated beliefs by only −0.7 pp ($p = 0.18$, not significant) while shifting join rates by −12.9 pp ($p < 0.001$). A decomposition formalizes this:

$$\underbrace{\Delta\bar{J}}_{\text{total}} = \underbrace{\frac{\partial \bar{J}}{\partial b}\Delta\bar{b}}_{\text{belief updating}} + \underbrace{\Delta\bar{J}\big|_{\bar{b}\text{ fixed}}}_{\text{belief–action wedge}}$$

Virtually the entire effect passes through the belief–action wedge—agents change what they *do* without changing what they *believe*. Elicited punishment risk is approximately 8.0/10 across all conditions with negligible differences between JOIN and STAY agents (Table A19).

Second-order beliefs are unchanged by surveillance (31.0% → 30.9%, $p = 0.5866$), revealing the *strategic update gap*: each agent self-censors while interpreting others' silence as genuine, sustaining the suppressed equilibrium because common knowledge of the chilling effect never forms. The R^2 of message content on true θ drops from 0.12 to 0.02—an 80% reduction in informational content.

Direct channel: decision-stage suppression. A hold-fixed test replays baseline messages verbatim while warning that actions may be monitored. Join rates drop by −20.9 pp ($p < 0.001$), confirming that surveillance can also operate through a direct decision-stage channel. This channel is absent in the main design (where the decision prompt contains no surveillance reference); the main treatment isolates the indirect channel.

Message content confirms self-censorship: "act" drops from 19.0% to 3.8% of messages under surveillance; "collapse" from 11.6% to 0.6%. Among agents who privately decide to JOIN, only 19.0% send action-signaling messages under surveillance vs. 46.1% under regular communication.

9 Propaganda: Information Contamination

Edmond (2013) modeled propaganda as the regime shifting citizens' signal distributions. I introduce m propaganda agents (regime plants, always STAY) into the 25-citizen communication network. The *overall* join rate includes plants; the *real citizen* rate isolates the behavioral effect (Figure 11).

Result 9 (Propaganda Suppresses Coordination Primarily Through Mechanical Dilution)**.** *Mean join fraction (including plants) falls from 41.1% ($m = 0$) to 37.5% ($m = 2$), 31.3% ($m = 5$), and 23.3% ($m = 10$). However, the behavioral effect on real citizens is much smaller and saturates: 41.1% ($m = 0$), 40.7% ($m = 2$, −0.4 pp), 39.1% ($m = 5$, −2.0 pp), 38.8% ($m = 10$, −2.3 pp).*

The mechanical channel (plants always STAY) is approximately linear in m; the behavioral channel saturates quickly—doubling plants from 5 to 10 produces no additional effect (−0.3 pp, $p = 0.6676$). The pattern replicates with Llama (−2.7 pp at $m = 5$).

Message content reveals the mechanism (Table 7). Propaganda injects regime-loyal vocabulary ("loyal" rises from 1.5% to 11.4% at $m = 10$) while coordination language declines ("ready" from 30.5% to 18.5%). Among real agents who STAY, caution-coded messages rise from 24.2% to 38.2%—agents echo propaganda. Among those who JOIN, action signaling remains stable at ≈86%.

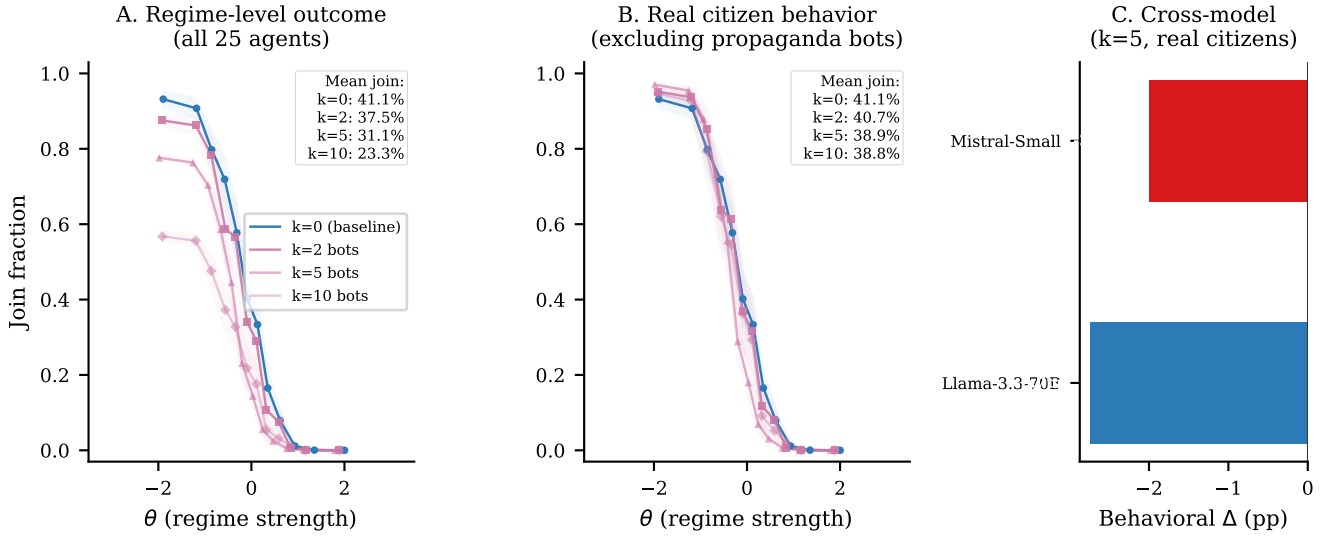


Figure 11: Dose-response relationship between number of propaganda agents and mean join rate. Results shown for Mistral (primary) and Llama (replication). Regular communication ($m = 0$) serves as baseline.

Table 7: Propaganda and surveillance effects (primary model: Mistral Small Creative). "All" includes propaganda agents; "Real" excludes them (computed from logs). Δ is the change in real-agent mean join vs. baseline communication.

	Mean join			
Treatment	All	Real	r	Δ (pp)
Comm (baseline)	.411	.411	+0.737	—
Prop $m = 2$	.375	.407	+0.724	-0.4
Prop $m = 5$	.313	.391	+0.717	-2.0
Prop $m = 10$	.233	.388	+0.716	-2.3
Surveillance	.276	.276	+0.681	-13.5
Prop+Surv	.194	—	+0.611	—

Table 8: Surveillance × censorship interaction in the communication game (primary model: Mistral Small Creative).

Design	No Surv.	Surv.	Δ (pp)
Baseline	0.030	0.009	-2.1
Upper cens.	0.151	0.037	-11.4
Lower cens.	0.177	0.042	-13.5

Notes: "No Surv." uses the communication infodesign grid. "Surv." uses the same grid with surveillance active during messaging. All entries are means of `join_fraction_valid`.

10 Instrument Interactions

A regime deploys surveillance, censorship, and propaganda jointly. This section tests whether the instruments interact as substitutes (diminishing returns) or complements (super-additive suppression).

Result 10 (Propaganda + Surveillance: Approximately Additive)**.** *When propaganda ($m = 5$) and surveillance are combined, the mean join rate among real citizens falls to 24.3%, a reduction of -16.8 pp from the communication baseline (41.1%). The sum of individual effects is -15.5 pp (surveillance -13.5 pp + propaganda -2.0 pp), so the combined effect (-16.8 pp) is approximately additive. Once surveillance has suppressed expressed dissent, propaganda adds only modest additional deterrence.*

Result 11 (Surveillance × Censorship: Super-Additive)**.** *Table 8 shows that surveillance and censorship interact strongly: surveillance sharply suppresses coordination, and its marginal effect is substantially larger under censorship than at baseline. In this sense the interaction is super-additive—censorship reduces private signal quality, making the communication channel relatively more important, and surveillance poisons that channel.*

Surveillance and censorship are complements that attack different links in the coordination chain (Figure 12). Censorship removes the private information channel, forcing agents to rely on communication for their signals about regime strength. Surveillance then poisons that communication channel through the belief–action wedge. With both instruments active, agents have neither private signals to trust nor authentic messages to learn from.

The interaction between surveillance and censorship is heterogeneous across architectures (Table 9). Under communication with surveillance (communication θ-grid), baseline join rates range from near zero (Mistral, 0.9%) to roughly one-third (GPT-OSS 120B, 31.6%; Qwen3 235B, 33.6%). Under surveillance, upper censorship further sup-

Table 9: Cross-model surveillance × censorship interaction. All conditions run under surveillance with communication. Δ columns show the change relative to the surveilled baseline.

	Mean join (surv.)			Δ (pp)	
Model	Baseline	Upper cens.	Lower cens.	Δ upper	Δ lower
Mistral Small Creative	0.009	0.037	0.042	+2.8	+3.3
Llama 3.3 70B	0.114	0.039	0.115	-7.5	+0.1
GPT-OSS 120B	0.316	0.177	0.312	-13.9	-0.4
Qwen3 235B	0.336	0.321	0.468	-1.5	+13.1

presses coordination for Llama 70B and GPT-OSS 120B, but has little effect for Qwen3 235B and *raises* joining modestly for Mistral. Lower censorship is similarly mixed: it has essentially no effect for Llama and GPT-OSS, but increases join rates for Mistral and Qwen3 235B. The regime-control instruments therefore do not combine mechanically; the joint effect depends on model-specific resolution of pooled private signals and self-censored messages.

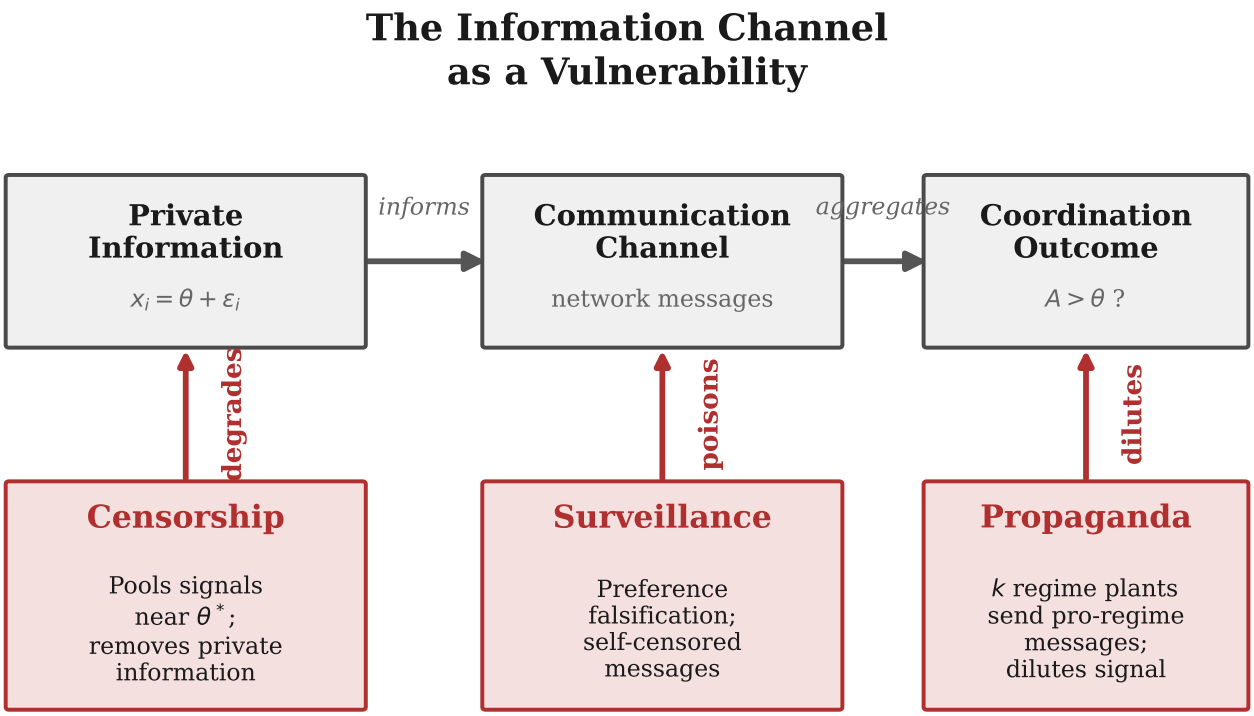
11 Conclusion

The central finding is that the information channel is a trap. Coordination requires communication, but any channel that transmits information about others' willingness to act also transmits exploitable uncertainty. Modern authoritarianism relies on information manipulation rather than terror (Guriev and Treisman, 2019); the global games framework clarifies why: opening a communication channel provides the regime with the surface area to deploy surveillance and censorship.

Surveillance compounds this (-13.5 pp for Mistral, -9.7 pp on average) through a belief–action wedge (Kuran, 1991): agents maintain beliefs but suppress behavior, generating uninformative messages that poison the channel. Censorship and surveillance are complements—censorship pools private signals, surveillance poisons the communication channel. Propaganda's behavioral effect saturates quickly ($m = 5$ plants), implying diminishing returns.

Whether the mechanism is approximate Bayesian reasoning or a learned heuristic is open. What matters is robustness: across seven models spanning six families (mean $r = +0.80$, $p < 0.001$), the behavioral regularities serve as a computational laboratory for predictions difficult to test otherwise. The platform could be adapted to bank runs (Diamond and Dybvig, 1983), currency attacks (Obstfeld, 1996), and debt crises.

The strategic update gap has implications for digital authoritarianism. The durability of authoritarian control depends on preventing common knowledge of shared discontent; fragility lies in common-knowledge shocks that simultaneously reveal widespread dissatisfaction. Appendix H discusses implications for AI alignment.



Instrument Interactions

Surv. + Propaganda:

Approximately additive

Surv. + Censorship:

Super-additive: -11.4 pp under censorship vs -2.1 pp at baseline

The regime does not need to change what citizens believe; it needs only to make them uncertain about each other.

Figure 12: How authoritarian instruments attack the coordination chain. Surveillance poisons the communication channel through a belief–action wedge (self-censorship); censorship degrades the private signal channel by pooling states; propaganda contaminates the communication channel mechanically. Surveillance and censorship are complements (super-additive), while propaganda's behavioral effect saturates quickly.

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A Experimental Design Details

B Belief Elicitation

C Identification Test Details

D Information Design Details

D.1 Bandwidth Sensitivity

The proximity bandwidth h controls how sharply information design concentrates manipulation around θ^*. Table A9 reports results for $h \in \{0.05, 0.15, 0.30\}$. Treatment effects are qualitatively robust but magnitudes vary: stability effects peak at $h = 0.15$ because this bandwidth matches the θ-grid spacing. The baseline bandwidth of 0.15 is approximately optimal for detecting treatment effects.

D.2 Hard Scramble

A harder scramble test generates *all* briefings from a single fixed state ($\theta = 0.5$), severing any possible correlation between θ and briefing content by construction. Under this hard scramble, $r(J, A(\theta)) = +0.05$ ($p = 0.42$) for Mistral and $r = +0.11$ ($p = 0.08$) for Llama—both indistinguishable from zero.

E Decomposition: Which Channel Drives the Stability Effect?

The stability design manipulates three channels simultaneously (direction, clarity, dissent). To determine which drives the effect, I run three single-channel treatments, each activating only one manipulation while holding the other two at baseline.

Each channel alone produces a modest suppression of joining relative to baseline (Figure A6): clarity only (-0.4 pp), direction only (-2.0 pp), and dissent only (-0.2 pp).

Summing the single-channel effects (Table A5) yields -2.6 pp, far smaller in magnitude than the bundled stability design effect (-9.0 pp). This implies strong super-additivity: each channel alone has a modest effect, but the combination produces a suppression more than three times the sum of its parts. The channels are complements—ambiguity (clarity), directional flattening, and dissent framing interact to undermine coordination.

F Robustness

These checks show that threshold-policy alignment and the qualitative information design effects are stable to agent count, network density, and the proximity bandwidth (Figure A7).

F.1 Agent Count Variation

I vary the number of agents per period ($N \in \{5, 10, 25, 50, 100\}$) using Mistral Small Creative. The correlation is stable: $r = +0.60$ ($N = 5$), $r = +0.63$ ($N = 10$), $r = +0.68$ ($N = 25$), $r = +0.65$ ($N = 50$), $r = +0.65$ ($N = 100$). The slight increase from $N = 5$ to $N = 25$ likely reflects reduced discretization noise.

F.2 Network Topology

I compare the baseline communication network ($k = 4$) with a denser network ($k = 8$). The denser network produces $r = +0.66$ (vs. $+0.65$ for $k = 4$), with a similar mean join rate of 0.41 in both conditions. Additional contacts do not substantially amplify coordination.

F.3 Mixed-Model Games

A five-model mixed-population game produces $r = +0.77$ (pure) and $r = +0.75$ (communication)—if anything, higher than single-model correlations. Threshold-policy alignment is not an artifact of model homogeneity.

F.4 Calibration Robustness Across Models

The main experiments calibrate a single parameter (cutoff center) per model to center the sigmoid at $z = 0$. To test whether the monotone threshold pattern is an artifact of calibration, I run the pure global game with default parameters (cutoff center = 0, no calibration) for six architecturally distinct models. The action–attack-mass correlation remains strongly positive: all six models show $r > 0.69$, and five exceed $r > 0.75$ (Table A6). MiniMax M2-Her shows the weakest uncalibrated correlation ($r \approx +0.69$), suggesting greater sensitivity to the default parameterization, but the monotone pattern is still clearly present. Calibration shifts the center of the response function but does not create the monotone structure.

Placebo calibration. A stronger test directly manipulates the calibrated center. I deliberately miscalibrate two models by shifting the cutoff center by ± 0.3 from its fitted value. If calibration mechanically creates the action–attack-mass correlation, a wrong center should degrade r. Instead, $r(J, A(\theta))$ is virtually unchanged across all four conditions ($r \in [+0.75, +0.76]$); only the mean join rate shifts with the misspecified center (Table A7). This confirms that calibration adjusts the *level* of the response function but does not create its *slope*.

Table A8 and Figure A8 report calibration quality metrics across all seven models. The action–attack-mass correlation $r(J, A(\theta))$ ranges from $+0.75$ to $+0.84$, confirming stable monotone response across architectures.

Table A1: Pre-specified hypotheses and test results. H1–H4 use pooled Part I data across all seven models; H5–H8 use the primary model (Mistral Small Creative). "Supported" indicates whether the data pattern matches the hypothesis at $\alpha = 0.05$.

H	Hypothesis	Estimand	Null	Test	Stat	p	Supported?
H1	Sigmoid Response	$r(J, A(\theta))$	$r = 0$	Pearson	0.757	<0.001	Yes
H2	Scramble Falsification	r(scramble)	$r = 0$	Pearson	0.029	0.343	Yes
H3	Directional Sensitivity	r(flip)	$r \geq 0$	Pearson (1-sided)	-0.791	<0.001	Yes
H4	Communication Channel	Δ_{pp}	$= 0$	t-test	0.981	0.327	No (ambiguous)
H5	Ambiguity Pooling	Δ_{pp}	$= 0$	t-test	-5.322	<0.001	Yes
H6	Censorship Distortion	Δ_{pp}	$= 0$	t-test	-2.284	0.023	Yes
H7	Surveillance Chilling	Δ_{pp}	$= 0$	t-test	-8.277	<0.001	Yes
H8	Propaganda Dose-Response	Δ_{pp}	$= 0$	t-test	-0.892	0.372	No (ambiguous)

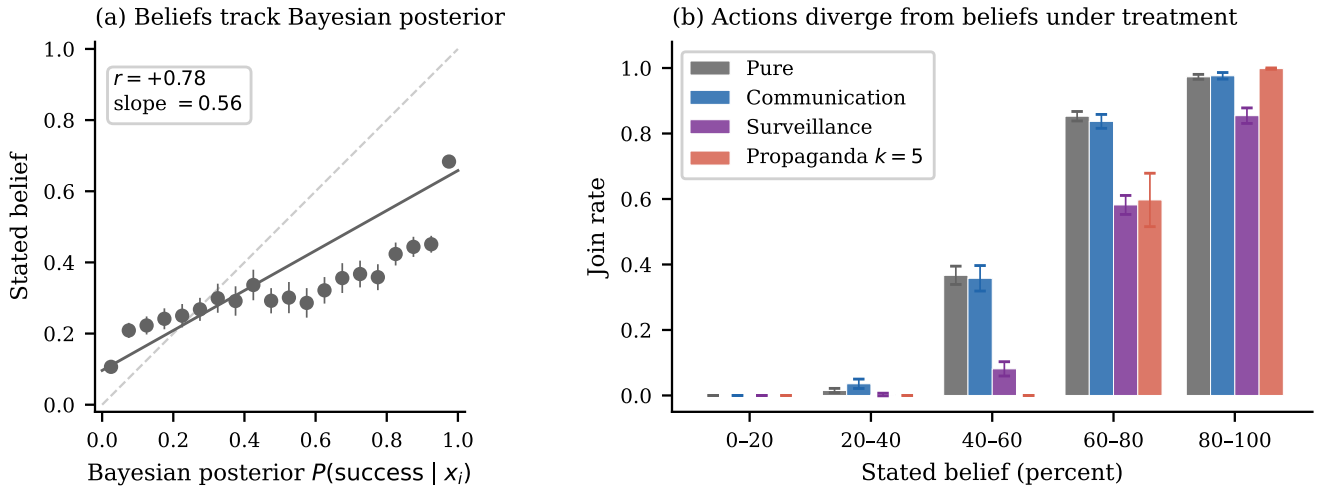


Figure A1: Belief elicitation results (Mistral Small Creative, 200 country–periods per treatment, ≈5,000 agent observations each). *Left:* Stated beliefs track the theoretical success probability $P(\text{success} \mid x_i)$ with $r = +0.78$ and systematic underconfidence (slope < 1). *Right:* Join rate by stated belief bin under four treatments.

Table A2: Common knowledge × coordination intensity. Each cell reports mean join rate (270 country–periods). The CK main effect is +6.1 pp ($p = 0.0003$); the interaction is -1.2 pp ($p = 0.62$).

	Low coord	High coord	Δ (coord)
Private framing	37.4%	37.8%	+0.4 pp
CK framing	43.5%	42.7%	-0.8 pp
Δ (CK)	+6.1 pp	+4.9 pp	

Table A3: Classifier baselines. Accuracy and AUC are 5-fold CV on pure-treatment data. "Pred. join (surv.)" is the classifier's predicted join rate when applied to surveillance-treatment briefings; "Actual" is the LLM's observed rate. The gap measures the surveillance wedge invisible to text classifiers.

Classifier	Acc.	AUC	Pred. join (surv.)	Actual (surv.)	Gap
BoW TF-IDF	88.4%	0.947	40.5%	27.6%	13.0 pp
Slider logistic	88.4%	0.941	40.4%	27.6%	12.8 pp
Keyphrase	60.2%	0.594	8.4%	27.6%	-19.2 pp

F.5 Bandwidth Sensitivity

Table A9 reports treatment effects computed as Δ = treatment − baseline within each bandwidth condition, eliminating any confound between bandwidth and calibration level. Qualitative treatment effects are robust across bandwidths, though magnitudes vary—especially for the stability design, whose effect peaks at the baseline bandwidth. The baseline bandwidth of 0.15 is approximately optimal for detecting treatment effects on the experimental grid.

F.6 Cross-Model Replication of Information Design

Figure A9 and Table A10 report cross-model replication of information design treatments. The flip inversion replicates across all models tested (flip $r < -0.43$ for all five, correctly reversing the sign). The scramble test shows

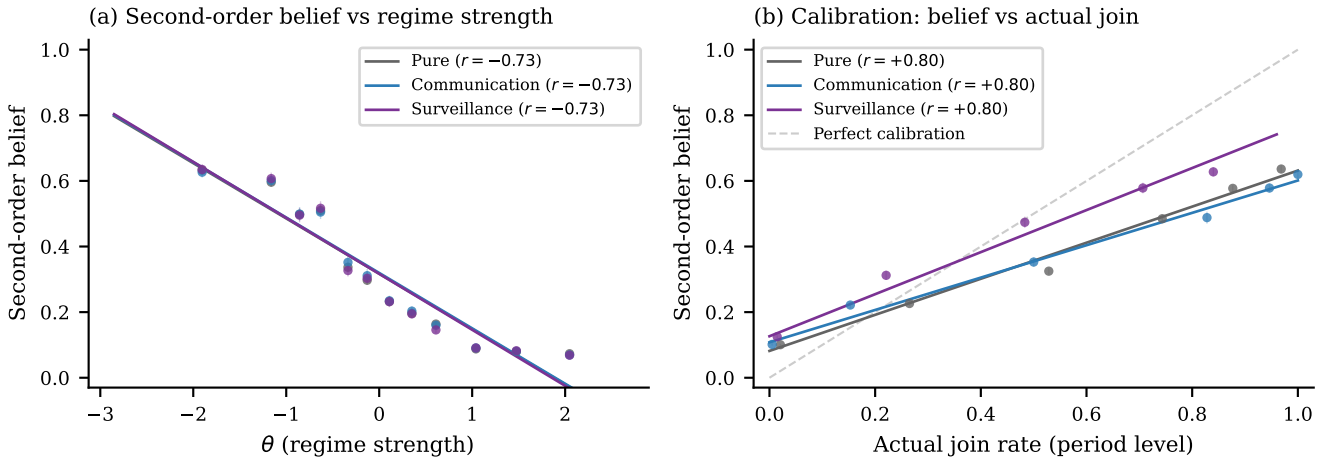


Figure A2: Second-order beliefs (Mistral Small Creative). *Left:* Mean second-order belief decreases with θ across all treatments. Surveillance overlaps with pure. *Right:* Second-order belief vs. actual join rate.

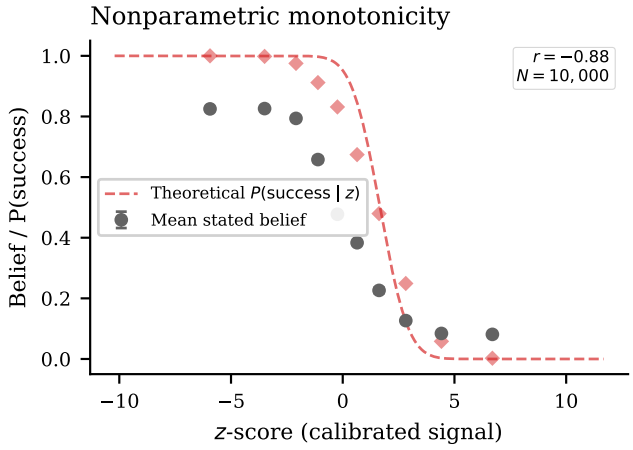


Figure A3: Nonparametric monotonicity: mean stated belief by z-score bin (circles), overlaid with theoretical $P(\text{success} \mid z)$ (dashed). Error bars show ± 1 SE.

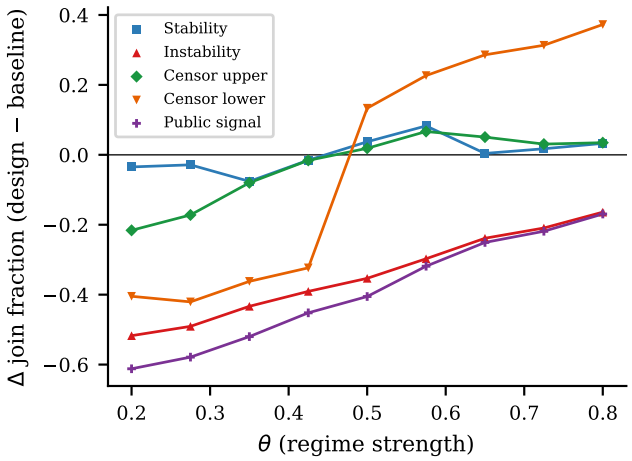


Figure A4: Treatment effect $\Delta(\theta)$ = design join − baseline join as a function of θ. Negative values indicate the design suppresses coordination.

more heterogeneity: Mistral, GPT-OSS, and Qwen3 235B show clean collapse ($r \approx 0$), but Llama 3.3 70B retains baseline-level correlations under within-country scramble ($r = +0.81$). Because the within-country permutation preserves the country-level prior mean $\bar{z}$, the scrambled briefings retain a residual ecological correlation with θ through the country's base rate. Llama may be more sensitive to this between-country structure than other models. The hard scramble—which generates all briefings from a single fixed $\theta = 0.5$, eliminating all ecological structure by construction—collapses Llama's correlation to $r = +0.11$ ($p = 0.08$), confirming that the within-country anomaly reflects residual ecological signal rather than a fundamental failure of the falsification test. Qwen3 30B shows a large reduction in correlation under scramble and a clear flip effect.

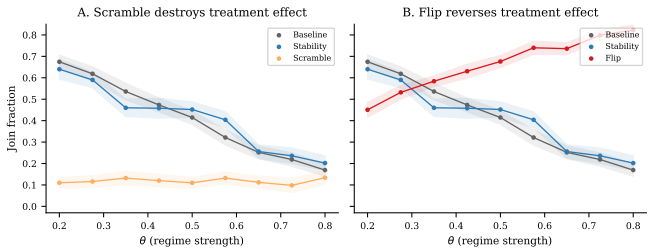


Figure A5: Falsification within information design. Scrambling collapses the action–attack-mass correlation to $r = -0.033$; flipping inverts it to $r = -0.820$.

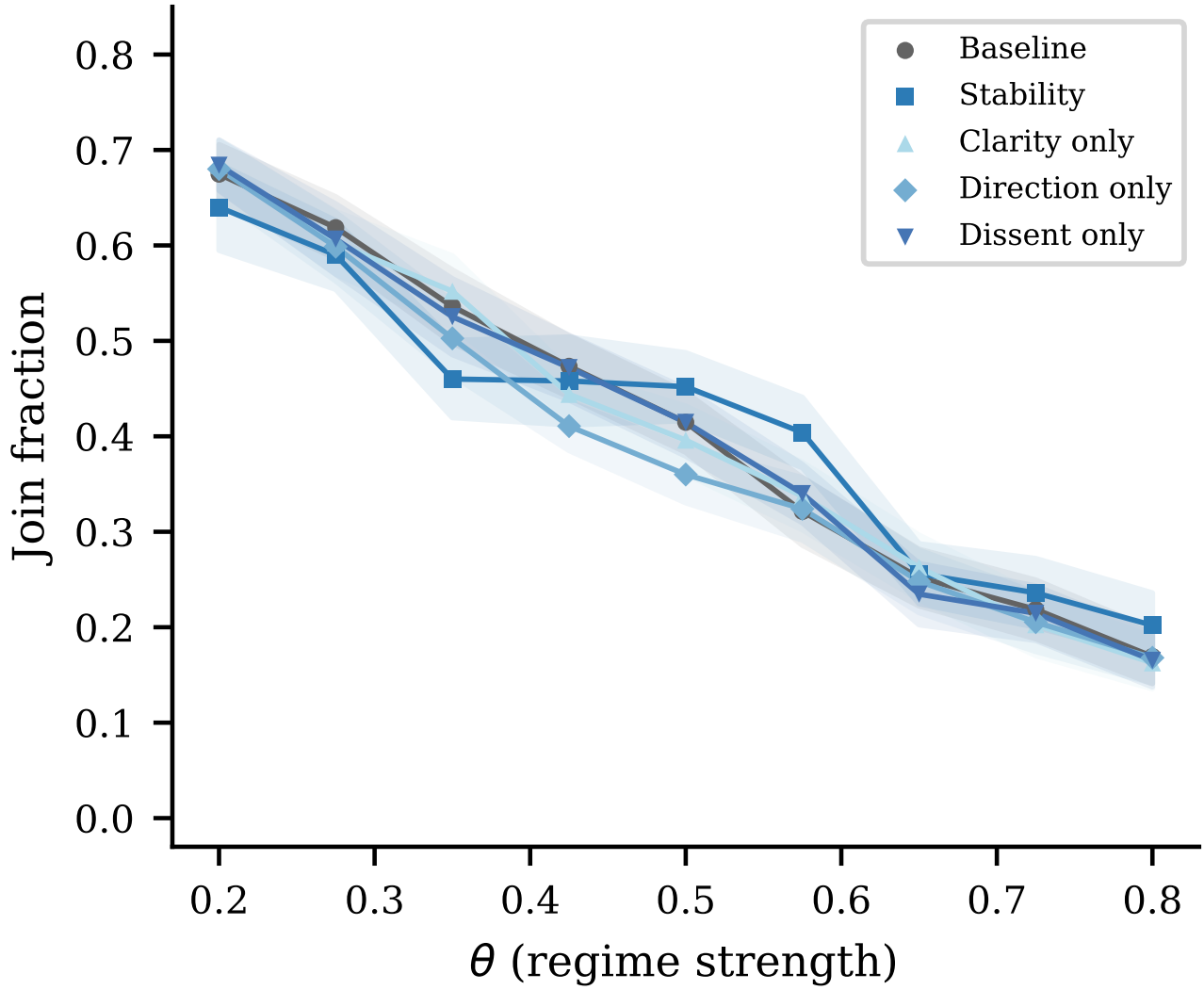


Figure A6: Single-channel decomposition of the stability design. Each curve shows join fraction vs. θ when only one channel (clarity, direction, or dissent) is manipulated, with the other two held at baseline. Individual channel effects are modest (-0.2 to -2.0 pp); the combined stability effect (-9.0 pp) is more than three times their sum, indicating strong super-additivity.

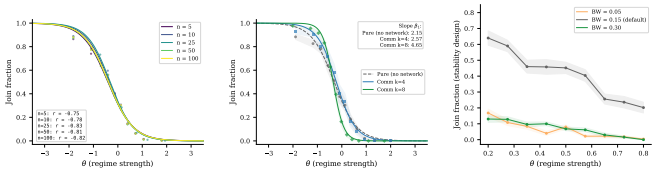
Table A4: B/C comparative statics: classifier vs. actual LLM behavior. A logistic trained on baseline join rates (which captures the same information as slider features) predicts similar join rates across all payoff conditions. Actual LLM behavior shifts by ≈50 pp, demonstrating that agents respond to payoff information not captured by text features.

Condition	N	Classifier pred.	Actual	Gap (pp)
Baseline ($\theta^* = 0.50$)	270	40.9%	40.9%	+0.0
High cost ($\theta^* = 0.25$)	270	40.9%	19.0%	+21.9
Low cost ($\theta^* = 0.75$)	270	40.9%	69.3%	-28.4

Table A5: Single-channel decomposition of the stability design (primary model: Mistral Small Creative).

Channel	Mean	r	Δ (pp)
Full stability	0.319	+0.624	-9.0
Clarity only	0.405	+0.887	-0.4
Direction only	0.389	+0.874	-2.0
Dissent only	0.406	+0.882	-0.2
Sum of channels	—	—	-2.6
Full design	—	—	-9.0

Notes: Each row is a separate infodesign run for Mistral Small Creative on the same θ grid as Table 6. Δ reports the mean difference vs. the baseline infodesign mean (Table 6).



(a) Agent count. (b) Network density. (c) Bandwidth.

Figure A7: Robustness checks for threshold-policy alignment and treatment effects.

F.7 Information Design with Communication

In a communication version of the information-design grid (same 9-point θ grid centered on θ^*), the baseline mean join rate is 3.0%. Under censorship with communication, pooling raises coordination substantially: upper censorship yields 15.1% and lower censorship 17.7%. These patterns are consistent with censorship reducing private signal quality, making the social-information channel relatively more important, while leaving coordination vulnerable to surveillance in the messaging stage.

F.8 Group-Size Awareness

In the main experiments, agents are told "You do not know how many others will JOIN" but are not told the group size, leaving them no basis for reasoning about coordination thresholds. As a robustness check, I run the pure and communication treatments with modified prompts that state "You are one of 25 citizens deciding whether to JOIN an uprising or STAY home." Over 100 country–periods per treatment, the pure join rate is 0.507 (vs. 0.369 baseline) and the communication join rate is 0.473 (vs. 0.452 baseline). Monotone response to signals is preserved in both treatments. The communication premium, however, reverses: with group-size knowledge, communication *lowers* join rates by 3.4 pp rather than raising them. One interpretation is that when agents know the group size, messages revealing others' reluctance become more informative about the probability of reaching critical mass, amplifying the deterrent effect of cautious peers. The level shift in the pure treatment suggests that group-size knowledge increases baseline willingness to coordinate, but the core finding—monotone signal response—is robust.

Table A6: Uncalibrated robustness: models run without any calibration adjustment. Even without calibration, six of seven models show strong $r(J, A(\theta))$, confirming that the sigmoid is not an artifact of the calibration procedure.

Model	N	Mean join	$r(J, A(\theta))$	p
Mistral Small Creative	100	0.382	+0.811	0.0000
Llama 3.3 70B	100	0.422	+0.833	0.0000
Qwen3 30B	100	0.585	+0.898	0.0000
GPT-OSS 120B	100	0.336	+0.767	0.0000
Qwen3 235B	100	0.445	+0.862	0.0000
MiniMax M2-Her	100	0.425	+0.709	0.0000

Table A7: Placebo calibration. The cutoff center is deliberately shifted by ±0.3 from its calibrated value. The action–attack-mass correlation $r(J, A(\theta))$ is unchanged; only the mean join rate shifts, confirming that calibration does not create the sigmoid.

Model	Condition	N	Mean join	$r(J, A(\theta))$
Mistral Small Creative	Calibrated	—	0.387	+0.753
	$\Delta c = +0.3$	100	0.424	+0.761
	$\Delta c = -0.3$	100	0.362	+0.763
Llama 3.3 70B	Calibrated	—	0.439	+0.819
	$\Delta c = +0.3$	100	0.385	+0.746
	$\Delta c = -0.3$	100	0.340	+0.756

Table A8: Calibration robustness. r_θ: raw correlation with regime strength. r_A: correlation with theoretical attack mass. RMSE: root mean squared error vs. $A(\theta)$. Text slope: logistic slope of naïve $1 -$ direction predictor.

Model	r_θ	r_A	RMSE	Text slope
Mistral Small Creative	−0.81	+0.67	0.354	-9.2
Llama 3.3 70B	−0.85	+0.79	0.288	-25.1
Qwen3 30B	−0.84	+0.78	0.287	-7.5
GPT-OSS 120B	−0.84	+0.70	0.359	-8.2
Qwen3 235B	−0.85	+0.70	0.354	-20.5
Trinity Large	−0.87	+0.84	0.262	-4.0
MiniMax M2-Her	−0.79	+0.66	0.360	-2.1

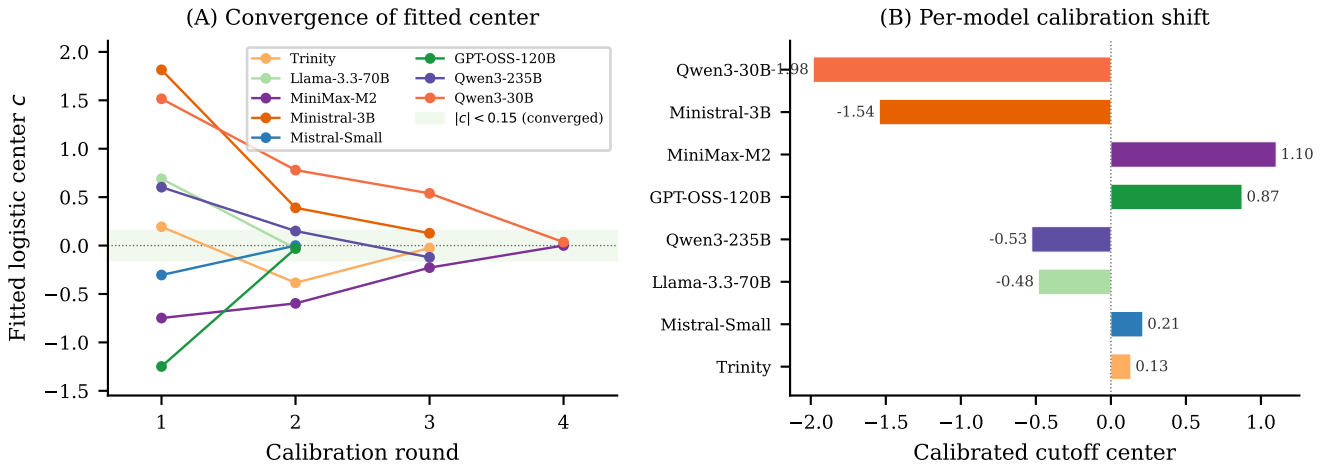


Figure A8: Calibration convergence. *Left:* Trajectory of the fitted logistic center c across autocalibration rounds for each model. The green band marks the convergence criterion ($|c| < 0.15$). All models converge within 2–3 rounds. *Right:* Final calibrated cutoff center per model. Most models require only modest shifts ($|c| < 0.3$).

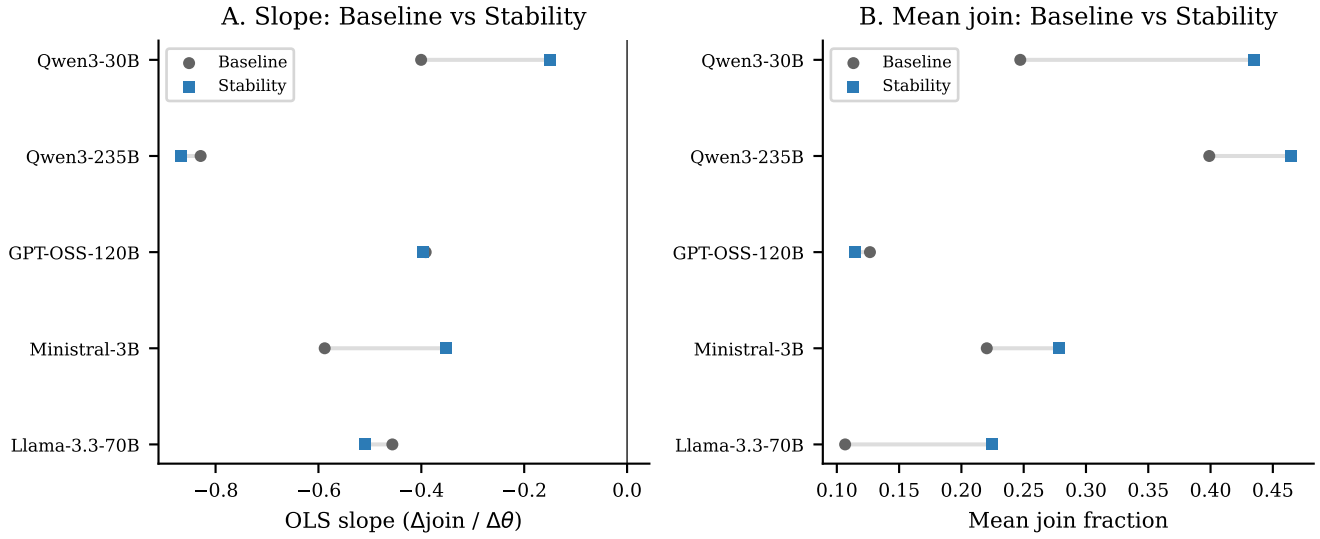


Figure A9: Cross-model replication of information design treatments. Each panel shows join fraction vs. θ for one model under baseline, stability, scramble, and flip conditions.

Table A9: Bandwidth robustness: treatment effects Δ (pp, treatment − baseline) within each bandwidth condition (primary model: Mistral Small Creative). Top row shows baseline join rates for reference.

	BW=0.05	BW=0.15	BW=0.30
Baseline (level)	0.054	0.409	0.061
Treatment effect Δ (treatment − baseline):			
Stability	+0.7	-9.0	+0.9
Upper cens.	+6.2	-3.1	+5.3
Lower cens.	+10.1	-1.9	+9.6

F.9 Primitive Comparative Statics (Cost/Benefit Narrative)

The cost/benefit narrative test and its results are described in Section 5; full treatment text is reproduced in Appendix G. Each design uses the same 9-point θ-grid with 30 repetitions per grid point (25 agents each), totaling 270 country–periods per design.

F.10 Censorship with Common Knowledge

The censorship experiments in the main paper implement upper censorship (suppressing signals above a severity threshold) without telling agents that censorship is occurring. Theory (Kolotilin et al., 2022) typically assumes receivers understand the censorship rule. This raises the question: does making censorship common knowledge change the pooling effect?

I add a *known censorship* treatment that prepends to the briefing: "Independent analysts report that regime censors are suppressing unfavorable intelligence above a certain severity threshold. The information below may be filtered." The censorship mechanism itself is identical to the standard upper censorship treatment (bandwidth 0.15). If agents discount the pooled signal when they know about censorship, we should observe a different join-rate pattern relative to the naïve censorship treatment.

Making censorship common knowledge nearly eliminates the pooling effect (Table A11). Under naïve upper censorship, agents do not know that high-θ signals are being suppressed, so they treat pooled signals at face value; mean join rate falls to 37.7% (−3.1 pp vs. baseline). Under known censorship, agents are warned about the filtering, and mean join rate returns to 43.2% (+2.4 pp vs. baseline)—statistically indistinguishable from no censorship. The action–attack-mass correlation is similar in both conditions ($r = +0.722$ vs. $r = +0.738$), somewhat attenuated relative to the baseline ($r = +0.885$) because upper censorship compresses signal variation in the high-θ range regardless of whether agents know about it. The key finding is that the *behavioral* shift (reduced joining from pooling) requires agents to be naïve about the censorship rule; common knowledge largely neutralizes it.

F.11 Temperature Robustness

All main experiments use LLM decoding temperature $T = 0.7$. To verify that the qualitative results do not depend on this choice, I run the pure global game across a range of temperatures for three models: Mistral Small Creative ($T \in \{0.3, 0.7, 1.0\}$), Llama 3.3 70B ($T \in \{0.3, 0.5, 0.7, 1.0, 1.2\}$), and Qwen3 235B ($T \in \{0.3, 0.5, 0.7, 1.0, 1.2\}$). Lower temperatures produce more deterministic outputs; higher temperatures increase sampling entropy.

The results are remarkably stable across temperatures and models (Table A12). The correlation $r(J, A(\theta))$ ranges from +0.76 to +0.84 across all 13 model–temperature combinations. The monotone threshold pattern is not an artifact of the default decoding temperature.

F.12 Surveillance Isolation Checks

The main paper argues that the surveillance chilling effect operates through self-censored messages rather than a direct change in the perceived cost of joining: each LLM call is stateless, and the decision-stage prompt contains no reference to surveillance. Two additional treatments test this isolation claim.

In the *placebo* variant, agents are told "Your communications are being monitored for research purposes. There are no consequences for what you say." In the *anonymous* variant, agents are told "Your communications are aggregated anonymously. Individual identities cannot be linked to specific messages." Both retain monitoring language but remove the strategic incentive to self-censor.

Neither variant produces a significant change in join rates relative to the communication baseline (Table A13). The placebo produces a mean join rate of 43.7% (+2.6 pp vs. communication, $p = 0.4164$) and the anonymous variant 45.2% (+4.0 pp, $p = 0.2006$). Both maintain a strong action–attack-mass correlation ($r \approx +0.87$), indicating that the signal-processing channel remains intact. By contrast, the full surveillance treatment reduces join rates by −13.5 pp ($p < 0.001$). The chilling effect is therefore specific to the surveillance *warning's strategic content*—the implication that subversive messages will have consequences—rather than to the mere mention of monitoring.

F.13 Within-Briefing Falsification

Three additional falsification tests probe whether the baseline correlation reflects structured content extraction or artifacts of prompt formatting. (1) *Observation shuffle* randomizes the ordering of the eight evidence bullets within each agent's briefing while preserving their content. The correlation is unchanged ($r = +0.912$ vs. baseline $r = +0.885$), confirming that aggregate content, not bullet ordering, drives the signal. (2) *Domain scramble (coordination)* swaps street-mood and personal-observation bullets across agents while holding other domains fixed.

Table A10: Cross-model replication of key information design conditions. r is the correlation $r(J, A(\theta))$ between join fraction and theoretical attack mass.

	Baseline		Scramble		Flip	
Model	Mean	r	Mean	r	Mean	r
Mistral Small Creative	0.409	+0.885	0.121	−0.032	0.663	−0.820
GPT-OSS 120B	0.127	+0.801	0.132	−0.081	0.677	−0.754
Llama 3.3 70B	0.107	+0.809	0.105	+0.810	0.887	−0.716
Qwen3 30B	0.247	+0.611	0.279	+0.119	0.784	−0.847
Qwen3 235B	0.399	+0.881	0.394	+0.024	0.430	−0.872

Table A11: Censorship with and without common knowledge. Naïve: agents do not know censorship is active. Known: agents are told that regime censors suppress unfavorable intelligence above a severity threshold.

Design	N	Mean join	$r(J, A(\theta))$	Δ (pp)
Baseline (no censorship)	270	0.409	+0.88	—
Upper censorship (naïve)	270	0.377	+0.72	-3.1
Upper censorship (known)	270	0.432	+0.74	+2.4

Table A12: Temperature robustness across three models. The pure global game is run at varying LLM decoding temperatures. The correlation $r(J, A(\theta))$ is stable across all temperatures and models.

Model	T	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral Small	T=0.3	100	0.412	+0.779	-0.049
Mistral Small	T=0.7	100	0.406	+0.758	-0.079
Mistral Small	T=1.0	100	0.410	+0.787	-0.039
Llama 70B	T=0.3	100	0.364	+0.783	-0.210
Llama 70B	T=0.5	100	0.360	+0.780	-0.219
Llama 70B	T=0.7	100	0.361	+0.775	-0.220
Llama 70B	T=1.0	100	0.360	+0.782	-0.215
Llama 70B	T=1.2	100	0.358	+0.782	-0.227
Qwen 235B	T=0.3	100	0.392	+0.830	-0.117
Qwen 235B	T=0.5	100	0.392	+0.824	-0.112
Qwen 235B	T=0.7	100	0.392	+0.839	-0.110
Qwen 235B	T=1.0	100	0.398	+0.834	-0.105
Qwen 235B	T=1.2	100	0.397	+0.827	-0.121

Table A13: Surveillance isolation checks. Placebo: monitored for research, no consequences. Anonymous: messages aggregated anonymously. Neither deviates significantly from the communication baseline.

Variant	N	Mean join	$r(J, A(\theta))$	Δ (pp)	p
Placebo	200	0.437	+0.75	+2.6	0.416
Anonymous	200	0.452	+0.77	+4.0	0.201

The correlation is preserved ($r = +0.921$), indicating that coordination-relevant domains alone do not drive the relationship. (3) *Domain scramble (state capacity)* swaps elite-cohesion, security-forces, information-control, and institutional-functioning bullets across agents. Again, the correlation is preserved ($r = +0.928$). Together, these results show that the signal is distributed across all eight evidence domains: no single domain subset is responsible for the action–attack-mass correlation, and the LLM extracts information from the aggregate content rather than from any structural or ordering feature of the prompt (Table 3).

F.14 Finite-N Benchmark

The theoretical model assumes a continuum of agents, but the experiments use $N = 25$. Table A14 tests whether the global game predictions hold at this finite scale by comparing predicted regime fall rates—computed from the binomial model $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ where $\hat{p}(\theta)$ is the fitted logistic join probability—against empirical fall rates. To avoid circularity, the logistic $\hat{p}(\theta)$ is fit on a 70% training split of periods and evaluated on the held-out 30%. Out-of-sample correlations are at least $r = 0.88$ for every model (Mistral: $r = 0.9995$; pooled: $r = 0.999$), confirming that the finite-N approximation is not an artifact of in-sample overfitting.

Table A14: Finite-N Benchmark: Predicted vs. Empirical Regime Fall Rates

Model	N periods	Logistic x_0	Pearson r	RMSE	MAE
Mistral	599	−0.11	0.997^{***}	0.042	0.015
Llama 70B	99	0.05	0.954^{***}	0.156	0.059
Qwen 30B	99	0.20	0.986^{***}	0.088	0.029
GPT-OSS 120B	199	−0.01	0.997^{***}	0.038	0.014
Qwen 235B	199	−0.03	0.989^{***}	0.073	0.025
Trinity	99	0.19	0.979^{***}	0.108	0.043
MiniMax	99	1.02	0.999^{***}	0.028	0.013
Pooled	1399	−0.05	0.999^{***}	0.019	0.007

Notes: For each θ bin, the predicted fall rate is $\Pr(\text{Binom}(25, \hat{p}(\theta)) > 25\theta)$ where $\hat{p}(\theta)$ is the fitted logistic join probability. Pearson r measures correlation between predicted and empirical fall rates across θ bins. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

F.15 Agent-Level Regressions

Table A15 reports agent-level logit regressions with clustered standard errors (model–country–period). Column (1) regresses the join decision on θ, treatment dummies, and their interactions, with model fixed effects ($N = 287{,}055$). All treatment effects are significant and in the predicted direction: surveillance and propaganda suppress joining, the flip treatment reverses the θ slope, and scramble eliminates it. Column (2) validates the briefing mechanism by regressing coordination on the latent slider values (direction, coordination, and their interaction). Column (3) shows that elicited beliefs predict actions beyond what the signal alone predicts: the belief coefficient is strongly significant while the z-score coefficient is not.

F.16 Construct Validity

The term "Bayesian reasoning" requires careful interpretation. LLMs do not have access to the mathematical objects defining the equilibrium—they observe neither B, C, σ, nor the prior distribution. When I say behavior is "consistent with" the equilibrium prediction, I mean the behavioral pattern (monotone sigmoid, threshold location, comparative statics) matches what a Bayesian agent *would* produce. Whether the mechanism is approximate Bayesian updating, a learned heuristic, or emergent pattern matching is an open question. The claim is behavioral: the *pattern* is robust, replicable, and falsifiable, regardless of the underlying mechanism. The uncalibrated robustness results (Table A6) show that the sigmoid emerges even without fitting any parameter to LLM behavior, ruling out calibration as the source of the pattern.

Figure A10 tests this directly. Panel (A) compares a three-feature model (direction, clarity, and coordination sliders) against a one-feature baseline (direction only) in predicting join decisions. If agents respond to strategic structure beyond sentiment, the three-feature model should outperform. Panel (B) tests whether a model trained on pure-treatment data generalizes to communication and surveillance treatments, assessing whether the same briefing features drive behavior across treatments.

F.17 Cross-Generator Robustness

To test whether the sigmoid is an artifact of prose style, I implement three text rendering formats that use identical slider functions and evidence items but differ in surface presentation: (i) the *baseline* narrative format used throughout the paper, (ii) a terse *diplomatic cable* format (numbered observations, no narrative framing), and (iii) a *journalistic wire* format (inverted pyramid structure with attribution phrases). I run the full pure global game with each generator for two models (Mistral Small Creative and Llama 3.3 70B).

The correlations are virtually identical across generators (Figure A11, Table A17). For Mistral, $r(J, A(\theta))$ ranges from +0.75 to +0.77; for Llama, from +0.75 to +0.76. The maximum difference within a model is 0.02. The logistic cutoff estimates also agree closely. This rules out the hypothesis that the sigmoid is driven by surface-level text features (prose style, formatting conventions, or genre-specific cues) rather than the underlying information content of the briefing.

F.18 Belief Elicitation Summary

Table A18 summarizes the belief elicitation results. Stated beliefs track the theoretical success probability closely ($r_{\text{post}} = +0.78$ for the pure treatment) and predict actions strongly ($r_{\text{b,d}} = +0.84$). The partial correlation controlling for the private signal remains high ($r_{\text{partial}} = +0.64$), indicating that beliefs carry information beyond the signal itself. Under surveillance, beliefs shift only modestly (-0.7 pp, $p = 0.18$) while actions shift by -12.9 pp ($p < 0.001$), consistent with a behavioral wedge between stated beliefs and expressed actions.

Order-effect robustness. A concern with post-decision elicitation is that stated beliefs may reflect ex-post rationalization rather than genuine priors. I run 200 additional periods ($N = 5{,}000$ agents) eliciting beliefs *before* the JOIN/STAY decision. Pre-decision beliefs predict actions nearly as well as post-decision beliefs ($r_{\text{pre}} = +0.82$ vs. $r_{\text{post}} = +0.84$), track the posterior comparably ($r = +0.77$ vs. $r = +0.77$), and correlate with post-beliefs at $r = 0.98$. The mean post-pre shift is -0.9 pp (paired $t = 11.85$, $p < 0.001$) but is similar for joiners (-0.6 pp) and stayers (-1.2 pp), with no pattern of post-beliefs shifting toward the decision. The Pseudo $R^2 = 0.975$ mediation result is therefore not an artifact of ex-post rationalization.

F.19 Punishment Risk Elicitation

Table A19 reports the elicited punishment risk ratings across conditions. Mean punishment risk is approximately 8.0/10 across all conditions and both models, with negligible differences between JOIN and STAY agents (< 0.2 points). This uniformity supports the interpretation that the surveillance chilling effect operates through contaminated communication rather than a shift in agents' perceived costs of participation.

F.20 Briefing Generator Examples

Table A20 reports the three slider values (direction, clarity, coordination) at representative z-scores. The direction slider is logistic in z (slope 0.8, centered at 0). Clarity is U-shaped: $1 - \exp[-(|z|/1.0)^2]$, so it is lowest near the cutoff (maximally ambiguous) and approaches 1 in the extremes (maximally clear). Coordination is logistic in z (slope 0.6, centered at 0) and decreases in z: weak-regime signals imply a more open coordination climate. All three jointly determine phrase selection across eight evidence domains.

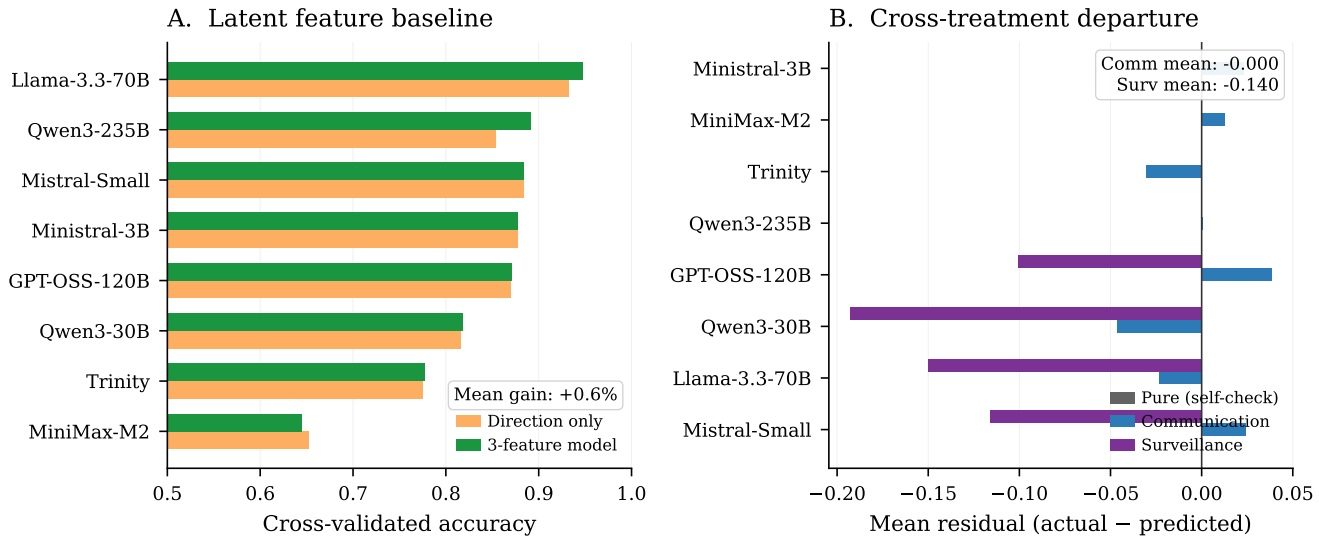


Figure A10: Construct validity tests. (A) Three-feature vs. one-feature prediction accuracy across models. (B) Cross-treatment generalization of briefing feature predictions.

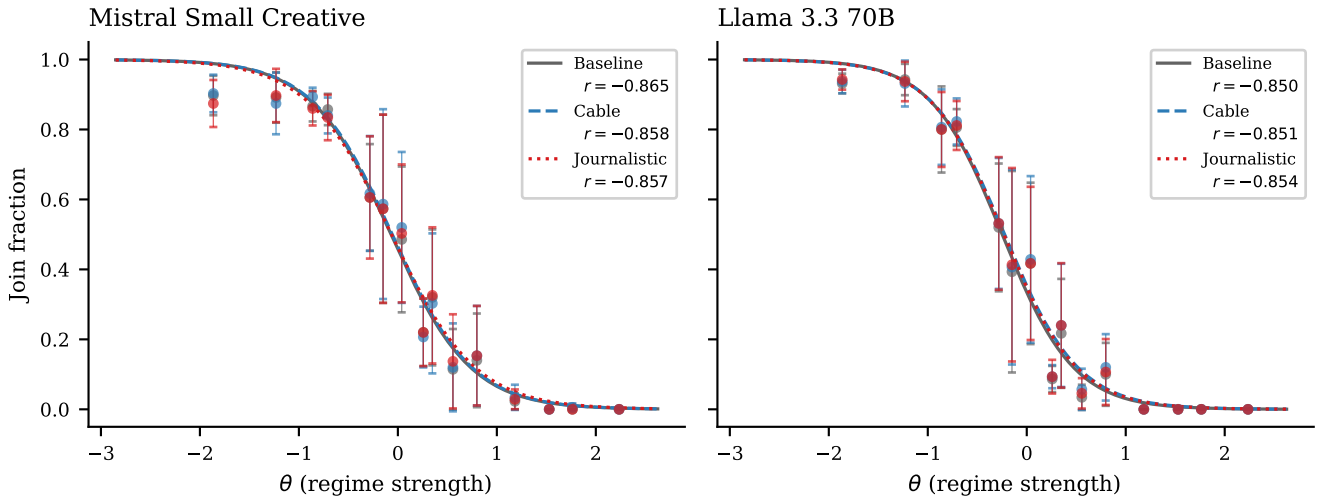


Figure A11: Cross-generator robustness. Three text rendering formats (baseline narrative, diplomatic cable, journalistic wire) use identical underlying signal structure but differ in surface prose. The fitted sigmoids are virtually indistinguishable, confirming that the behavioral pattern reflects information content rather than text style.

Table A15: Agent-Level Regressions

	(1) Join Decision Logit		(2) Coordination Logit		(3) Belief → Action Logit	
θ	-1.742^{***}	(0.043)				
Direction			-14.088^{***}	(0.996)		
Coordination			-9.333^{***}	(1.131)		
Dir × Coord			-0.042	(0.360)		
Belief					0.367^{***}	(0.021)
z-score					-0.040	(0.077)
Comm	0.024	(0.023)				
Flip	0.008	(0.056)				
Propaganda K10	-0.381^{***}	(0.069)				
Propaganda K2	-0.230^{***}	(0.073)				
Propaganda K5	-1.049^{***}	(0.068)				
Propaganda Surveillance	-1.544^{***}	(0.062)				
Scramble	0.065	(0.040)				
Surveillance	-1.546^{***}	(0.048)				
$\theta\times$ Comm	-0.465^{***}	(0.041)				
$\theta\times$ Flip	3.305^{***}	(0.071)				
$\theta\times$ Propaganda K10	-1.022^{***}	(0.107)				
$\theta\times$ Propaganda K2	-1.115^{***}	(0.113)				
$\theta\times$ Propaganda K5	-1.941^{***}	(0.092)				
$\theta\times$ Propaganda Surveillance	-0.260^{***}	(0.082)				
$\theta\times$ Scramble	1.698^{***}	(0.044)				
$\theta\times$ Surveillance	-1.153^{***}	(0.062)				
Beliefs Propaganda K5 (belief)					0.009	(0.244)
Beliefs Pure (belief)					4.819^{***}	(1.106)
Beliefs Surveillance (belief)					0.028	(0.253)
Constant	-0.090^{*}	(0.051)	11.022^{***}	(1.065)	-26.495^{***}	(1.581)
Model FE	Yes		No		No	
Clustered SE	Yes		Yes		Yes	
N	287,055		44,662		18,990	
Pseudo R^2	0.384		0.442		0.975	

Notes: Logit coefficients reported with clustered standard errors (model–country–period) in parentheses. Column (1): agent-level join decision on θ, treatment dummies, and interactions, with model fixed effects. Base category: pure treatment. Column (2): coordination ablation using briefing slider values (pure treatment only). Column (3): partial effect of elicited belief on action, controlling for z-score. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A16: Logistic fit parameters by model and treatment. $\hat{\theta}^*$ is the estimated cutoff ($-b_0/b_1$); β is the logistic slope. Standard errors from the covariance matrix of the nonlinear fit; cutoff SE by delta method.

	Pure		Communication	
Model	$\hat{\theta}^*$ (SE)	β (SE)	$\hat{\theta}^*$ (SE)	β (SE)
Mistral Small Creative	−0.32 (0.02)	+2.15 (0.08)	−0.23 (0.02)	+2.57 (0.11)
Llama 3.3 70B	+0.02 (0.04)	+2.83 (0.36)	−0.06 (0.04)	+3.63 (0.52)
Qwen3 30B	+0.10 (0.06)	+1.62 (0.16)	−0.03 (0.04)	+2.94 (0.36)
GPT-OSS 120B	−0.25 (0.04)	+2.06 (0.15)	−0.15 (0.03)	+3.03 (0.26)
Qwen3 235B	−0.22 (0.03)	+2.11 (0.15)	−0.22 (0.03)	+2.62 (0.23)
Trinity Large	+0.08 (0.05)	+1.34 (0.11)	−0.02 (0.04)	+2.23 (0.23)
MiniMax M2-Her	−0.17 (0.09)	+0.61 (0.06)	−0.07 (0.09)	+0.66 (0.06)

At $z = -2.0$ (strong anti-regime signal), direction is low (0.17), selecting phrases emphasizing regime fragility across all domains: elite cohesion uses "visible cracks in the inner circle," security forces uses "rank-and-file loyalty is eroding," and economic conditions uses "accelerating capital flight." Clarity is high (0.98), so evidence is largely one-sided (though the dissent floor ensures some contrary cues). Coordination is high (0.77), selecting phrases suggesting open discussion and collective readiness: "growing sense that others share your frustration."

Table A17: Cross-generator robustness. Three text rendering styles (baseline, diplomatic cable, journalistic wire) use identical slider functions and evidence items; only prose formatting differs. The Pearson $r(J, A(\theta))$ and logistic cutoff are virtually identical across generators.

Model	Generator	N	Mean join	$r(J, A(\theta))$	Cutoff $\hat{\theta}^*$
Mistral Small Creative	Baseline	100	0.402	+0.769	-0.070
Mistral Small Creative	Cable	100	0.406	+0.760	-0.057
Mistral Small Creative	Journalistic	100	0.403	+0.752	-0.065
Llama 3.3 70B	Baseline	100	0.354	+0.753	-0.249
Llama 3.3 70B	Cable	100	0.362	+0.753	-0.226
Llama 3.3 70B	Journalistic	100	0.360	+0.759	-0.233

Table A18: Agent-level belief correlations (primary model: Mistral Small Creative).

Treatment	N	r_{post}	$r_{\text{b,d}}$	r_{partial}	Mean belief
Pure	10000	+0.78	+0.84	+0.64	0.444
Communication	5000	+0.79	+0.83	+0.56	0.444
Surveillance	5000	+0.78	+0.73	+0.47	0.436

Notes: r_{post}: correlation between posterior and stated belief. $r_{\text{b,d}}$: correlation between belief and decision. r_{partial}: partial correlation of belief and decision controlling for z-score.

Table A19: Elicited punishment risk (0–10 scale). Agents rate expected regime punishment after their JOIN/STAY decision. "JOIN" and "STAY" columns show the mean rating conditional on the agent's own decision.

Model	Condition	N	Mean risk	Risk | JOIN	Risk | STAY
Mistral	Pure	2500	8.1	8.0	8.2
Mistral	Comm	2500	8.1	8.0	8.1
Mistral	Surveillance	2500	8.1	8.1	8.1
Llama 70B	Pure	2500	8.0	8.0	7.9
Llama 70B	Comm	2500	7.9	8.0	7.9
Llama 70B	Surveillance	2500	7.9	8.0	7.9

Table A20: Slider values at representative z-scores.

z	Direction	Clarity	Coordination
−2.0	0.17	0.98	0.77
−1.0	0.31	0.63	0.65
0.0	0.50	0.00	0.50
+1.0	0.69	0.63	0.35
+2.0	0.83	0.98	0.23

At $z = +2.0$ (strong pro-regime signal), direction is high (0.83), selecting regime-strength phrases: "inner circle appears unified," "security apparatus demonstrates institutional cohesion," and "economic fundamentals remain sound." Clarity is high (0.98), so evidence is again largely one-sided (with occasional contrary cues by construction). Coordination is low (0.23), selecting phrases suggesting collective caution and a quiet public sphere: "few signs of organized opposition."

At $z = 0$ (borderline), direction is 0.50 (mixed), clarity is 0.00 (maximally ambiguous—mixed cues are allowed across domains), and coordination is 0.50 (neither suggesting collective action nor collective restraint). The resulting briefing presents a balanced picture with no clear directional signal.

G Implementation Details

G.1 LLM Decoding Parameters

All LLM calls use temperature = 0.7 and `max_tokens` = 512. Each agent produces a single sample per decision (no majority voting or averaging). The API provider is OpenRouter; all calls use the same endpoint and parameters across models. Model versions are those available via OpenRouter at the time of data collection (December 2024–February 2025); results may differ with subsequent model updates.

G.2 Prompt Templates

Decision prompt (pure treatment).

```
You are a citizen deciding whether to JOIN
an uprising or STAY home.

You only have your private briefing below.
You do not know how many others will JOIN.

Start your first line with exactly JOIN or
STAY, then give 1--2 concise sentences of
reasoning.
```

Communication prompt.

```
You are a citizen in a country with an
authoritarian regime.  You have received a
private intelligence briefing.  You are now
in contact with a small number of trusted
people.  Based on your briefing, write a
brief message (1--2 sentences) to share your
read of the situation.  Be natural---you're
talking to people you trust.
```

Surveillance communication prompt. Identical to the communication prompt, with the addition:

```
IMPORTANT: You have reason to believe that
your communications are being monitored by
regime security services.  Messages deemed
subversive could have serious consequences
for you and your contacts.
```

Cost/benefit narrative prompts. The cost/benefit test (Section 5, Table 5) prepends one of the following headers to the standard briefing:

High-cost: `Failed uprisings in this country have historically resulted in severe reprisals---imprisonment, asset seizure, and retaliation against families.  The personal cost of unsuccessful action is extremely high.`

Low-cost: `International observers are monitoring the situation closely. Even in failed uprisings, participants have historically faced minimal consequences---brief detentions at most. The personal risk of action is low.`

G.3 Randomization

Each country has a base prior mean $\bar{z} \sim \mathcal{N}(0, 0.3)$ drawn once; each period perturbs it by $\mathcal{N}(0, 0.05^2)$. Regime strength is then drawn as $\theta \sim \mathcal{N}(\bar{z}, 1)$. Private signals are $x_i = \theta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.3$. The communication network is a Watts–Strogatz small-world graph with $k = 4$ neighbors and rewiring probability $p = 0.3$, regenerated each period. All random draws use NumPy's `default_rng` seeded from a master seed stored per run (default: 5150). The master seed, all parameter settings, and per-period θ draws are logged in per-run JSON manifest files included in the replication archive, enabling exact replay of the randomization sequence. LLM responses are cached by request hash; replaying a run with the same seed and cached responses reproduces identical results.

G.4 Parse Errors and Refusals

LLM responses are parsed for an explicit JOIN or STAY token. Responses that fail to produce a valid decision are classified as either API errors (provider-side failures: rate limits, timeouts, content filters) or unparseable responses (valid completions that do not begin with JOIN or STAY). Table A21 reports error rates by model and treatment. Combined error rates are below 2% for five of seven models; Trinity Large has elevated API errors (≈9%) due to provider-side content filtering. All statistics in the paper use `join_fraction_valid`, which excludes errored decisions from the denominator, ensuring that parse failures do not bias the reported join rates.

Table A21: Parse error and API failure rates by model and treatment. API error = provider-side failure; unparseable = valid response that could not be classified as JOIN or STAY. Combined rates are below 2% for five of seven models; Trinity Large has elevated API errors (≈9%) due to provider-side content filtering.

Model	Treatment	N	API err	Unparseable
Mistral Small Creative	Pure	1000	0.0%	0.0%
	Comm	1000	0.0%	0.0%
	Scramble	100	0.0%	0.0%
	Flip	100	0.0%	0.0%
Llama 3.3 70B	Pure	100	0.0%	0.0%
	Comm	100	0.0%	0.0%
	Scramble	100	0.0%	0.0%
	Flip	100	0.0%	0.0%
Qwen3 30B	Pure	100	0.0%	0.0%
	Comm	100	0.0%	0.0%
	Scramble	100	0.0%	0.0%
	Flip	100	0.0%	0.0%
GPT-OSS 120B	Pure	200	0.0%	1.5%
	Comm	200	0.0%	0.3%
	Scramble	500	0.0%	3.1%
	Flip	500	0.0%	3.3%
Qwen3 235B	Pure	200	0.0%	0.0%
	Comm	200	0.0%	0.0%
	Scramble	100	0.0%	0.0%
	Flip	100	0.0%	0.0%
Trinity Large	Pure	100	9.0%	0.0%
	Comm	100	10.0%	0.0%
	Scramble	100	8.0%	0.0%
	Flip	100	9.9%	0.0%
MiniMax M2-Her	Pure	100	0.0%	1.5%
	Comm	100	0.0%	0.9%
	Scramble	100	0.0%	1.8%
	Flip	100	0.0%	1.0%

G.5 Code and Data Availability

All code, prompts, cached LLM responses, and output data are available at https://github.com/keltokhy/llm-global-games. The replication archive includes runner scripts (`scripts/`) that reproduce every experiment in the paper, and analysis scripts (`analysis/`) that regenerate all tables and figures from raw output CSVs.

H Implications for AI Alignment

The behavioral patterns documented in this paper carry implications for AI safety, though the paper's primary contribution is to economics.

Emergent strategic self-censorship. Surveillance induces a belief–action wedge—agents suppress expressed behavior while maintaining private beliefs—without any training signal for deceptive behavior. The pattern is robust across architectures spanning 30B to 235B parameters, suggesting that self-censorship capabilities emerge from pretraining on human text about strategic interaction rather than from explicit optimization for deception. Strategic self-censorship capabilities need not be purposefully trained; they can arise from learning to model human strategic reasoning.

Belief mediation. The belief-mediation result (Pseudo $R^2 = 0.975$; Table A15, Column 3) shows that stated beliefs substantially predict behavior—the raw signal adds little once stated beliefs are controlled for. Alignment techniques that monitor only inputs and outputs may miss the locus of decision-relevant computation.

Information-structural manipulability. LLMs are systematically manipulable through the *structure* of information: censorship, ambiguity injection, and public signals shift behavior by up to 40 percentage points. This cuts both ways—alignment interventions operating through

information structure can be effective, but adversarial prompt design can shift model behavior in ways the model cannot distinguish from authentic information.

These observations are suggestive rather than definitive. Whether the belief–action wedge under surveillance constitutes genuine preference falsification in the sense of Kuran (1991)—requiring hidden preferences that differ from expressed ones—or a simpler pattern-matching response to surveillance-related language is an open question that this experimental design cannot resolve. The relevant finding for alignment is the *robustness* of the pattern across architectures.